

Econometric Model and Estimation - Overview

Economic theory suggests a relationship

$$y = f(x_1, x_2, \dots, x_k)$$

Parametric vs nonparametric model

Parametric model - functional form is known, but the parameters of the function are unknown

Linear vs nonlinear parametric models

Nonparametric model - functional form is unknown

Semiparametric and seminonparametric models, Partial linear model

Estimation Methods of Parametric Models

Least squares method (OLS and GLS), Minimum Absolute Difference (MAD) method, maximum likelihood (ML) method, method of moment (MM) and generalized method of moment (GMM), maximum empirical likelihood (MEL) method

Estimation Methods of Nonparametric Models - Kernel method, Nadaraya-Watson kernel method

Statistical Inferences

Classical methods: Likelihood ratio test, Wald test, Lagrange Multiplier (LM) test or Score test

Standard Linear Regression Model (SLM)

Linear Regression Model

A linear regression model specifies the dependent variable (regressand) y_t as a sum of a linear function of the K observable explanatory variables (regressors) X_{tk} , $k=1,2,\dots,K$, and an unobservable error term u_t ,

$$y_t = X_t\beta + u_t, \quad t=1,2,\dots,N$$

where the subscript t indicates the t^{th} observation, X_t is a K -dimensional row vector and β is a K -dimensional column vector of unknown coefficients. This is written in a matrix form for the entire N observations as

$$y = X\beta + u$$

where the dimensions of the matrices are $y:N \times 1$, $X:N \times K$, $\beta:K \times 1$, and $u:N \times 1$.

Ordinary Least Squares (OLS) Estimation method

The OLS estimation method does not require information about the statistical properties of the variables in the model. It just needs to specify a linear (or nonlinear) regression equation. Properties of the OLS estimator of course depends on the statistical properties of the variables.

The least squares method of estimation of unknown coefficients β_k finds the vector $\hat{\beta}$ that minimizes the sum of squared residuals (SSR)

$$\min_{\beta} SSR = \sum_{t=1}^N (y_t - X_t\beta)^2 = (y - X\hat{\beta})'(y - X\hat{\beta})$$

Taking derivatives of the SSR with respect to β we find the first and the second order conditions

$$\frac{\partial SSR}{\partial \beta} = -2X'y + 2X'X\hat{\beta} = 0, \quad \frac{\partial^2 SSR}{\partial \beta \partial \beta'} = 2X'X > 0$$

The "ordinary" least squares estimator (OLSE) of β is the solution of the first order condition $X'X\hat{\beta} = X'y$, which is called the "normal equations." The OLS predictor of the dependent variable is defined as

$$\hat{y} = X\hat{\beta}$$

Properties of the OLSE

The OLSE is the best linear unbiased estimator (BLUE) under certain assumptions. We present below the minimum set of assumptions for each property of the OLSE.

Assumption 1: $\text{rank}(X)=K$ and $\text{rank}(X, y)=K+1$

The assumption of $\text{rank}(X)=K$ implies that the columns of X (observation vectors of each regressor) are linearly independent and the number of observations N is greater than K . This assumption is called the assumption of full column rank. The assumption of $\text{rank}(X, y)=K+1$ implies that there is no exact solution for the unknown coefficient β in a linear equation $y=X\beta$. If this rank is K , then y is linearly dependent on

the columns of X and we can find a nonzero vector β such that $y = \sum_{k=1}^K \beta_k X_k$, where X_k is the k-th column of X. Under these assumptions, the OLSE has the following properties.

- (a) $\hat{\beta}$ is **unique** and is given by $\hat{\beta} = (X'X)^{-1}X'y$. This is because $X'X$ is positive definite (p.d.) due to the assumption of full column rank of X.
- (b) $\hat{\beta}$ is a **linear** estimator, i.e., a linear function of y ($\hat{\beta}_k = b_{k1}y_1 + b_{k2}y_2 + \dots + b_{kN}y_N$, $k=1,2,\dots,K$)
- (c) Let the OLS regression residual vector be $\hat{u} = y - X\hat{\beta} = y - \hat{y}$. Then,
 - (c.1) $\hat{u} = My = Mu$, where $M = I - X(X'X)^{-1}X'$ is a symmetric idempotent matrix with rank N-K. That is, $M = M'$ and $MM = M$.
 - (c.2) $\hat{u}$ is orthogonal to the columns of X, i.e., $X_k'\hat{u} = \sum_{t=1}^N X_{tk}\hat{u}_t = 0$ for all k, and hence $X'\hat{u} = 0$. This means that if one regresses residuals $\hat{u}$ against regressors X again, the coefficients will be identically zero.
 - (c.3) If X has a column of a constant, then $\sum_t \hat{u}_t = 0$ and $\bar{y} = \bar{X}\hat{\beta}$, where $\bar{y}$ and $\bar{X}$ are the samples means of y and X. That is, the regression line passes through the point of sample means. If X does not have a column of a constant, the regression line does not pass through the point of sample means.

Assumption 2: $E(u|X) = 0$

This assumption implies that the regressors are strongly exogenous, i.e., $E(u_t|X_s) = 0$ for all t and s. This implies that $E(y|X) = X\beta$. The weak exogeneity of X requires only $E(u_t|X_t) = 0$ for all t. If X is a fixed variate, then X is strongly exogenous. Under this additional assumption

- (d) The OLSE $\hat{\beta}$ is **unbiased**, i.e., $E(\hat{\beta}|X) = \beta$ for all β .
- (e) Let R be a J×K matrix of nonzero constants. An unbiased estimator of linearly estimable linear parametric functions $R\beta$ is given by $R\hat{\beta}$.
- (f) The residual $\hat{u}$ is an unbiased estimator of the error term u in the sense that $E(\hat{u} - u|X) = 0$.

Assumption 3: $E(uu'|X) = \sigma^2 I$

This assumption implies that the error terms (u_t) are homoscedastic (have the same variance) and uncorrelated with each other conditional on X.

$$E(u_t^2|X_s) = \sigma^2 \text{ for all } t, s$$

$$\text{cov}(u_t, u_{t'}|X_s) = 0 \text{ for all } t \neq t', s$$

With this assumption added to the previous assumptions, we have the following key results for the OLSE.

- (g) $V \equiv \text{cov}(\hat{\beta}|X) = \sigma^2(X'X)^{-1}$ and $\text{cov}(R\hat{\beta}|X) = \sigma^2 R(X'X)^{-1}R' = RVR'$ for a matrix R of constants.
- (h) **Gauss-Markov Theorem:** The OLSE $\hat{\beta}$ is the unique BLUE of β under assumptions (1)-(3).

- (i) The unique BLUE of linearly estimable linear parametric functions $R\beta$ is given by $R\hat{\beta}$.
- (j) $\text{cov}(\hat{u}|X) = \sigma^2 M$, which is singular. Thus, $\hat{u}_t$ and $\hat{u}_s$ are correlated ($\text{cov}(\hat{u}_t, \hat{u}_s | X_s) \neq 0$) even if we assumed that the true error terms are uncorrelated ($\text{cov}(u_t, u_{t'} | X_s) = 0$ for $t \neq t'$).
- (k) Let the LSE of σ^2 be
- $$\hat{\sigma}^2 = SSR/(N-K) = \hat{u}'\hat{u}/(N-K)$$
- Then, $\hat{\sigma}^2$ is an unbiased estimator of σ^2 , and hence, the unbiased estimator of $\text{cov}(\hat{\beta}|X)$ is $\hat{V} = \hat{\sigma}^2(X'X)^{-1}$, and the unbiased estimator of $\text{cov}(R\hat{\beta}|X)$ is $R\hat{V}R' = \hat{\sigma}^2 R(X'X)^{-1}R$.

Proof.

- (g) Omitting the conditioning expression for notational simplicity, we can write

$$\begin{aligned} \text{cov}(\hat{\beta}) &= E[(\hat{\beta} - E(\hat{\beta}))(\hat{\beta} - E(\hat{\beta}))'] = E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)'] = E[(X'X)^{-1}X'u u'X(X'X)^{-1}] \\ &= (X'X)^{-1}X' E(uu')X(X'X)^{-1} = (X'X)^{-1}X'(\sigma^2 I)X(X'X)^{-1} = \sigma^2(X'X)^{-1} \end{aligned}$$

where the third equality follows from

$$\hat{\beta} = (X'X)^{-1}X'y = (X'X)^{-1}X'(X\beta + u) = \beta + (X'X)^{-1}X'u.$$

The covariance of $R\hat{\beta}$ is derived in a similar way. Since $E(R\hat{\beta}) = R\beta$,

$$\text{cov}(R\hat{\beta}) = E[R\hat{\beta} - R\beta][R\hat{\beta} - R\beta]' = E[R(\hat{\beta} - \beta)(\hat{\beta} - \beta)'R'] = RE[(\hat{\beta} - \beta)(\hat{\beta} - \beta)']R' = \sigma^2 R(X'X)^{-1}R'$$

- (h) Linearity and unbiasedness of the OLSE are shown above. To show that the OLSE is the best estimator among all linear unbiased estimators β^* , we need to show that $\text{cov}(\beta^*) - \text{cov}(\hat{\beta})$ is positive semidefinite under the assumptions (1)-(3). Let $\beta^* = Ay = AX\beta + Au$ be any other linear estimator for a known matrix A . Unbiasedness of β^* requires $AX = I$, because $E(\beta^*) = AX\beta$ under the assumption $E(u|X) = 0$. The covariance of the linear unbiased estimator β^* is $\text{cov}(\beta^*) = \sigma^2 AA'$. Hence, we only need to show that $AA' - (X'X)^{-1}$ is positive semidefinite. It is easy to verify by using $AX = I$ that

$$AA' - (X'X)^{-1} = [A - (X'X)^{-1}X'] [A' - X(X'X)^{-1}] \equiv CC'$$

which is always positive semidefinite.

- (i) Let β^* be as defined above. Then,

$$\text{cov}(R\beta^*) - \text{cov}(R\hat{\beta}) = \sigma^2 RAA'R' - \sigma^2 R(X'X)^{-1}R' = \sigma^2 R[AA' - (X'X)^{-1}]R' = \sigma^2 (RCC'R') \equiv \sigma^2 DD'$$

which is positive semidefinite.

Assumption 4: $(u|X)$ is distributed as a multivariate normal $(0, \sigma^2 I)$

This assumption implies that the distribution of y conditional on X is a multivariate normal $N(X\beta, \sigma^2 I)$. With this added assumption, we can specify the distribution functions of $\hat{\beta}$ and $\hat{\sigma}^2$:

- (l) $\hat{\beta} \sim N(\beta, V)$, $V = \sigma^2(X'X)^{-1}$, and $\hat{u} \sim N(0, \sigma^2 M)$, $M = I - X(X'X)^{-1}X'$
- (m) $W \equiv (N-K)\hat{\sigma}^2/\sigma^2 = \hat{u}'\hat{u}/\sigma^2 = u'Mu/\sigma^2 \sim \chi^2(N-K)$
- (n) $\hat{\beta}$ is independent of $\hat{\sigma}^2$

These results can be proven easily by using Theorem 2(i) for (l), Theorem 3(iv) for (m) and Theorem 4(i) for

(n) in the **Statistical Appendix**.

Let R be a $J \times K$ matrix of known constants with rank $J \times K$, and q be a $J \times 1$ vector of known constants. Then, results in (l)-(n) imply (See the Theorems in Statistical Appendix)

$$(o) \quad R\hat{\beta} \sim N(R\beta, RVR')$$

$$(p) \quad Z^* \equiv (R\hat{\beta} - R\beta)'(RVR')^{-1}(R\hat{\beta} - R\beta) \sim \chi^2(J)$$

$$(q) \quad Z \equiv (R\hat{\beta} - q)'(RVR')^{-1}(R\hat{\beta} - q) \sim \chi^2(J; \delta), \quad \delta^2 = (R\beta - q)'(RVR')^{-1}(R\beta - q)$$

$$(r) \quad F^* \equiv \frac{Z^*/J}{W/(N-K)} = \frac{1}{J} (R\hat{\beta} - R\beta)'(R\hat{V}R')^{-1}(R\hat{\beta} - R\beta) \sim F(J, N-K), \text{ where } \hat{V} = \hat{\sigma}^2(X'X)^{-1}$$

$$(s) \quad F \equiv \frac{Z/J}{W/(N-K)} = \frac{1}{J} (R\hat{\beta} - q)'(R\hat{V}R')^{-1}(R\hat{\beta} - q) \sim F(J, N-K; \delta)$$

When the number of restriction is 1, we need to consider the following t -statistic. Let $Z_1 \sim N(\mu, 1)$ and $Z_2 \sim \chi^2(m)$. If Z_1 and Z_2 are independent, then

$$t = \frac{Z_1}{\sqrt{Z_2/m}} \sim t(m, \mu)$$

where μ is the noncentrality parameter which can be either positive, or zero, or negative. If $\mu=0$, it is a central t distribution.

(t) If $J=1$ and $R\hat{\beta}$ is a scalar, its true variance and estimated variance are also scalars:

$$\text{var}(R\hat{\beta}) = \sigma^2 R(X'X)^{-1}R' = RVR',$$

$$\text{est. var}(R\hat{\beta}) = \hat{\sigma}^2 R(X'X)^{-1}R' = R\hat{V}R'$$

$$\text{sd}(R\hat{\beta}) = [\sigma^2 R(X'X)^{-1}R']^{1/2} = (RVR')^{1/2},$$

$$\text{est. sd}(R\hat{\beta}) = [\hat{\sigma}^2 R(X'X)^{-1}R']^{1/2} = (R\hat{V}R')^{1/2}$$

$$t^* = \frac{(R\hat{\beta} - R\beta)/\text{sd}(R\hat{\beta})}{\sqrt{W/(N-K)}} = \frac{(R\hat{\beta} - R\beta)/\text{sd}(R\hat{\beta})}{\sqrt{\hat{\sigma}^2/\sigma^2}} = \frac{(R\hat{\beta} - R\beta)}{\text{est. sd}(R\hat{\beta})} \sim t(N-K)$$

$$t = \frac{(R\hat{\beta} - q)}{\text{est. sd}(R\hat{\beta})} \sim t(N-K, R\beta - q)$$

For example, consider a row vector $R = (0, 0, \dots, 0, 1, 0, \dots, 0)$ which has 1 in the k^{th} position. Then, $R\hat{\beta} = \hat{\beta}_k$, and $R\hat{V}R' = \hat{v}_{kk}$ is the k^{th} diagonal element of $\hat{V}$, which is the estimate of the variance of $\hat{\beta}_k$. Therefore, we can write

$$t^* = \frac{\hat{\beta}_k - \beta_k}{\sqrt{\hat{v}_{kk}}} \sim t(N-K), \quad t = \frac{\hat{\beta}_k - q}{\sqrt{\hat{v}_{kk}}} \sim t(N-K, \beta_k - q)$$

For another example, consider a row vector $R = (a, b, 0, \dots, 0)$ where a and b are constants. Then, $R\hat{\beta} = a\hat{\beta}_1 + b\hat{\beta}_2$, and $R\hat{V}R' = a^2\hat{v}_{11} + b^2\hat{v}_{22} + 2ab\hat{v}_{12}$, which is the estimated variance of $R\hat{\beta}$, and hence,

$$t^* = \frac{(a\hat{\beta}_1 + b\hat{\beta}_2) - (a\beta_1 + b\beta_2)}{\text{est. } SD(a\hat{\beta}_1 + b\hat{\beta}_2)} \sim t(N-K)$$

$$t = \frac{(a\hat{\beta}_1 + b\hat{\beta}_2) - q}{\text{est. } SD(a\hat{\beta}_1 + b\hat{\beta}_2)} \sim t(N-K, a\beta_1 + b\beta_2 - q)$$

Remark: It is easy to verify the following relationship between t distribution and F distribution:

$$t(N-K)^2 \sim F(1, N-K)$$

For the first example,

$$t^{*2} = \frac{(\hat{\beta}_k - \beta_k)^2}{\hat{v}_{kk}} = (\hat{\beta}_k - \beta_k)'(\hat{v}_{kk})^{-1}(\hat{\beta}_k - \beta_k) = F^* \sim F(1, N-K)$$

Measurement of Goodness of Fit

How well do the regressors explain the dependent variable? How do we measure the goodness of the regression? A measure of the goodness of fit is the squared correlation coefficient between the dependent variable y_t and the least squares predictor $\hat{y}_t = X_t\beta$:

$$R^2 \equiv [\text{corr}(y, \hat{y})]^2 = \frac{[\text{cov}(y, \hat{y})]^2}{\text{var}(y) \text{var}(\hat{y})} = \frac{[(y - \bar{y})'(\hat{y} - \bar{\hat{y}})/N]^2}{[(y - \bar{y})'(y - \bar{y})/N][(\hat{y} - \bar{\hat{y}})'(\hat{y} - \bar{\hat{y}})/N]}$$

where $\bar{y}$ and $\bar{\hat{y}}$ are the vectors of sample means of y_t and $\hat{y}_t$, respectively. Since R^2 is a squared correlation coefficient, it takes a value between 0 and 1 ($0 \leq R^2 \leq 1$), and its value is independent of the measurement unit. A high R^2 is interpreted as a good fit and the regressors explain the variation of the dependent variable well. This expression can be simplified further by using the following results.

Lemma. When a constant is one of the regressors,

- (a) $\bar{y} = \bar{\hat{y}}$
- (b) $\hat{u}'(\hat{y} - \bar{y}) = 0$
- (c) $(y - \bar{y})'(\hat{y} - \bar{y}) = (\hat{y} - \bar{y})'(\hat{y} - \bar{y})$
- (d) $(y - \bar{y})'(y - \bar{y}) = (\hat{y} - \bar{y})'(\hat{y} - \bar{y}) + \hat{u}'\hat{u}$

Substituting these results into the equation, R^2 can be rewritten as

$$R^2 = \frac{(\hat{y} - \bar{y})'(\hat{y} - \bar{y})}{(y - \bar{y})'(y - \bar{y})} = 1 - \frac{\hat{u}'\hat{u}}{(y - \bar{y})'(y - \bar{y})}$$

This is the expression more commonly used to define R^2 , which is called the *coefficient of (multiple) determination*. The denominator term is called the *total sum of squares* (SST) and is a measure of the variation of the dependent variable. The numerator in the first expression is called the *explained sum of squares* or *regression sum of squares* (SSR), and the SSR $\hat{u}'\hat{u}$ is also called the *unexplained sum of squares* or *error sum of squares* (SSE).

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

Note that SSR in this notation is different from the sum of squared residuals. (notations are indeed confusing)

A serious drawback of using R^2 as a measure of goodness of fit is that it is a nondecreasing, and typically an increasing, function of the number of regressors. One can increase the value of R^2 by simply ‘throwing in’ more variables into the regression equation, even if they do not belong to the equation in theory. To avoid this problem in assessing the goodness of the regression equation, the *adjusted* R^2 (adjusted for the degrees of freedom) is often used.

$$\bar{R}^2 = 1 - \frac{(\hat{u}'\hat{u})/(N-K)}{(y - \bar{y})'(y - \bar{y})/(N-1)} = 1 - \frac{N-1}{N-K}(1 - R^2), \quad -\frac{K-1}{N-K} \leq \bar{R}^2 \leq 1$$

Note that $\bar{R}^2$ is less than 1, but it can take a negative value if $R^2 < (K-1)/(N-K)$.

Model Selection - AIC and BIC

Consider two alternative models

$$(1) y = X_1\beta_1 + u_1, \quad (2) y = X_2\beta_2 + u_2$$

where X_i has K_i variables and $K_1 < K_2$. One may choose one of these models based on their values of $\bar{R}^2$, choosing one that has a higher $\bar{R}^2$. This is equivalent to choose a model that has a smaller value of $1 - \bar{R}^2$:

$$(1) \quad 1 - \bar{R}_1^2 = \frac{N-1}{N-K_1}(1-R_1^2) = \frac{N-1}{N-K_1} \frac{\hat{u}_1' \hat{u}_1}{SST} = \frac{N-1}{SST} \frac{1}{N-K_1} \hat{u}_1' \hat{u}_1 \propto \frac{1}{N-K_1} \hat{u}_1' \hat{u}_1$$

$$(2) \quad 1 - \bar{R}_2^2 = \frac{N-1}{N-K_2}(1-R_2^2) = \frac{N-1}{N-K_2} \frac{\hat{u}_2' \hat{u}_2}{SST} = \frac{N-1}{SST} \frac{1}{N-K_2} \hat{u}_2' \hat{u}_2 \propto \frac{1}{N-K_2} \hat{u}_2' \hat{u}_2$$

Since $(N-1)/SST$ does not depend on the number of regressors, we need to compare only the last expressions. They can be interpreted as penalized sum of squared residuals where the penalty is $1/(N-K_j)$. The size of penalty is greater for a model with a larger number of regressors is, i.e., $1/(N-K_1) < 1/(N-K_2)$. To summarize, the model selection based on the adjusted R-squared is equivalent to a model selection based on the penalized SSR. This criterion function can also be written in a logarithmic form

$$\ln(1 - \bar{R}_j^2) \propto \ln\left(\frac{\hat{u}_j' \hat{u}_j}{N}\right) + \ln\left(\frac{1}{N-K_j}\right)$$

There are other criterion functions that are more popular: the Akaike information criterion (AIC) and the Bayesian Information criterion (BIC), which is also called the Schwartz information criterion (SIC). They differ in the specification of the penalty function

$$\begin{aligned} \text{AIC:} \quad & \ln\left(\frac{\hat{u}_j' \hat{u}_j}{N}\right) + 2\frac{K_j}{N} \\ \text{BIC:} \quad & \ln\left(\frac{\hat{u}_j' \hat{u}_j}{N}\right) + \ln(N)\frac{K_j}{N} \end{aligned}$$

We will have a chance to discuss these criteria more later.

The Best Linear Unbiased Predictor

Consider a linear regression model $y_1 = X_1\beta + u_1$ and its OLS estimator $\hat{\beta}$ which is the BLUE under the assumptions (A1)-A(3). Suppose we wish to predict the value of dependent variable at a new values of regressors X_2 when the regression relationship has not changed, i.e., $y_2 = X_2\beta + u_2$ and $u = (u_1', u_2')'$ satisfies the assumptions (A1)-A(3).

Theorem. The best linear unbiased predictor (BLUP) of y_2 for a given X_2 is given by $\hat{y}_2 = X_2\hat{\beta}$.

Proof. (i) Linearity. $\hat{y}_2$ is a linear function of y_1 because $\hat{\beta}$ is a linear function of y_1 .

(ii) Unbiasedness. The prediction error $e_2 = y_2 - \hat{y}_2$ has a zero mean, because

$$E(e_2) = E[X_2(\beta - \hat{\beta}) + u_2] = X_2(\beta - \hat{\beta}) = 0$$

(iii) Best. The prediction error has the smallest covariance matrix. To show this,

$$\begin{aligned} \text{cov}(e_2) &= E[e_2 - E(e_2)][e_2 - E(e_2)]' = E[u_2 - X_2(\hat{\beta} - \beta)][u_2 - X_2(\hat{\beta} - \beta)]' \\ &= \sigma^2 I_{N_2} + X_2 \text{cov}(\hat{\beta}) X_2' = \sigma^2 I_{N_2} + \sigma^2 X_2 (X_1' X_1)^{-1} X_2' \end{aligned}$$

Let another linear predictor be $\tilde{y}_2 = A y_1$ and prediction error $\tilde{e}_2 = y_2 - \tilde{y}_2$. We need to choose A such that the predictor is unbiased ($E(\tilde{e}_2) = 0$). And then we need to show $\text{cov}(\tilde{e}_2) - \text{cov}(\hat{e}_2)$ is a positive semidefinite. The homework question asks you prove this.

Homework #1

- (1) What is the objective function of the least squares estimation of a linear parametric model?
- (2) Prove the statements (c.1)-(c.3) listed above.
- (3) What is the minimum set of assumptions for an OLSE to be unique? to be unbiased? to be BLUE?
- (4) Prove (d)-(f), (j) and (k) listed above.
- (5) Prove the statements in the Lemma in the section of goodness of fit.
- (6) When one of the regressors is a constant, show that the two expressions for R^2 are the same.

$$R^2 = [\text{corr}(y_p, \hat{y}_p)]^2 = \frac{[\text{cov}(y_p, \hat{y}_p)]^2}{\text{var}(y_p) \text{var}(\hat{y}_p)} = \frac{[(y - \bar{y})'(\hat{y} - \bar{\hat{y}})/N]^2}{[(y - \bar{y})'(y - \bar{y})/N][(\hat{y} - \bar{\hat{y}})'(\hat{y} - \bar{\hat{y}})/N]}$$

$$R^2 = 1 - \frac{\hat{u}'\hat{u}}{(y - \bar{y})'(y - \bar{y})}$$

- (7) Suppose your estimation equation does not include an intercept (constant) term, and yet you compute R^2 by the second expression. What do you expect to happen to the value of R^2 ?
- (8) Explain why an additional regressor in the equation will always increase (or at least will not reduce) R^2 , regardless of its relevance as an explanatory variable for the dependent variable.
- (9) Consider the theorem about the BLUP. Prove the last part of bestness.

LSE of SLM with Non-stochastic Linear Restrictions on Coefficients

Suppose we wish to estimate a SLM

$$y = X\beta + u$$

subject to a set of linear restrictions on coefficient

$$R\beta = q$$

where R is a $J \times K$ matrix of known constants with a full row rank, $\text{rank}(R)=J$, and q is a $J \times 1$ vector of known constants.

Let λ be a J -dimensional column vector of Lagrange multipliers. Then, the Lagrange equation is

$$\mathcal{L} = (y - X\beta)'(y - X\beta) + 2\lambda'(R\beta - q)$$

and the first order conditions

$$\partial \mathcal{L} / \partial \beta = -2X'y + 2X'X\beta + 2R'\lambda = 0$$

$$\partial \mathcal{L} / \partial \lambda = 2(R\beta - q) = 0$$

Rearranging the terms, these conditions can be written in a matrix form as

$$\begin{pmatrix} X'X & R' \\ R & 0 \end{pmatrix} \begin{pmatrix} \beta \\ \lambda \end{pmatrix} = \begin{pmatrix} X'y \\ q \end{pmatrix} \Rightarrow A\delta = d \Rightarrow \hat{\delta} = A^{-1}d$$

It is easy to verify that A is nonsingular if $|X'X| \neq 0$, and its inverse can be found by applying the inverse of a partitioned matrix:

$$\begin{pmatrix} X'X & R' \\ R & 0 \end{pmatrix}^{-1} = \begin{pmatrix} (X'X)^{-1} - (X'X)^{-1}R'C^{-1}R(X'X)^{-1} & (X'X)^{-1}R'C^{-1} \\ C^{-1}R(X'X)^{-1} & -C^{-1} \end{pmatrix}$$

where $C = R(X'X)^{-1}R'$. The restricted least squares estimators of β and λ are then derived from $A^{-1}d$:

$$\tilde{\beta} = \hat{\beta} - (X'X)^{-1}R'C^{-1}(R\hat{\beta} - q) = \hat{\beta} - VR'S^{-1}(R\hat{\beta} - q)$$

$$\tilde{\lambda} = C^{-1}(R\hat{\beta} - q) = \sigma^2 S^{-1}(R\hat{\beta} - q)$$

where

$$V = \text{cov}(\hat{\beta}) = \sigma^2(X'X)^{-1}$$

$$S = \text{cov}(R\hat{\beta}) = R[\sigma^2(X'X)^{-1}]R' = RVR'$$

Properties of the restricted LSE

(a) Moments of $\tilde{\beta} = \hat{\beta} - VR'S^{-1}(R\hat{\beta} - q)$

$$(a.1) E(\tilde{\beta}) = \beta - VR'S^{-1}(R\beta - q)$$

$$(a.2) \tilde{V} \equiv \text{cov}(\tilde{\beta}) = V - VR'S^{-1}RV$$

$$(a.3) \text{cov}(\hat{\beta}) - \text{cov}(\tilde{\beta}) = V - \tilde{V} = VR'S^{-1}RV \succeq 0$$

$$(a.4) \text{MSE}(\hat{\beta}) - \text{MSE}(\tilde{\beta}) = VR'S^{-1}[S - (R\beta - q)(R\beta - q)']S^{-1}RV \geq 0$$

(a.1) shows that $\tilde{\beta}$ is unbiased *if and only if* the restrictions are correct ($R\beta = q$). (a.2) shows that $\tilde{\beta}$ has a singular covariance matrix since there are only $K-J$ independent parameters. This is seen by noting $R\tilde{V} = 0$. (a.3) indicates that the variance of each $\tilde{\beta}_k$, or the variance of any linear combination (e.g. $a\tilde{\beta}_1 + b\tilde{\beta}_3$), is less than or equal to the variance of corresponding unrestricted estimator $\hat{\beta}_k$ or linear combination. Note, however, that this does not imply that the *estimated cov*($\hat{\beta}$) is greater than the *estimated cov*($\tilde{\beta}$), because the restricted estimate $\tilde{\sigma}^2$ is different from the unrestricted estimate $\hat{\sigma}^2$. (a.4) provides for an easy example that a biased estimator can have a smaller mean squared error (MSE) if the reduction in variance at the expense of the bias is sufficiently large. Proofs of these results are straightforward. For (a.4), one has to only note

$$\text{MSE}(\hat{\beta}) = E(\hat{\beta} - \beta)(\hat{\beta} - \beta)' = \text{cov}(\hat{\beta}) = V$$

$$\text{MSE}(\tilde{\beta}) = E(\tilde{\beta} - \beta)(\tilde{\beta} - \beta)' = V - VR'S^{-1}RV + VR'S^{-1}(R\beta - q)(R\beta - q)'S^{-1}RV$$

(b) Moments of $\tilde{\lambda} = \sigma^2 S^{-1}(R\hat{\beta} - q)$.

$$(b.1) E(\tilde{\lambda}) = \sigma^2 S^{-1}(R\beta - q)$$

$$(b.2) \text{cov}(\tilde{\lambda}) = \sigma^4 S^{-1}$$

$$(b.3) \text{cov}(\tilde{\beta}, \tilde{\lambda}) = 0$$

Note that $E(\tilde{\lambda}) = 0$ if and only if the restrictions are true ($R\beta - q = 0$).

(c) Properties of restricted regression residuals $\tilde{u} = y - X\tilde{\beta}$.

$$(c.1) \tilde{u} = y - X\tilde{\beta} = \hat{u} + XVR'S^{-1}(R\hat{\beta} - q) = \tilde{M}u + XVR'S^{-1}(R\beta - q)$$

$\tilde{M} = M + XVR'S^{-1}R(X'X)^{-1}X'$ is a symmetric idempotent matrix of rank $(N-K+J)$

$$M = I - X(X'X)^{-1}X'$$

$$(c.2) E(\tilde{u}) = XVR'S^{-1}(R\beta - q), \quad \text{cov}(\tilde{u}) = \sigma^2 \tilde{M}$$

$$(c.3) \text{cov}(\tilde{u}, \tilde{\beta}) = 0, \quad \text{cov}(\tilde{u}, \tilde{\lambda}) = 0$$

(d) Properties of $SSR_r \equiv \tilde{u}'\tilde{u}$

$$(d.1) SSR_r = SSR_u + \sigma^2(R\hat{\beta} - q)'S^{-1}(R\hat{\beta} - q) = u'\tilde{M}u + 2u'\tilde{M}XVR'S^{-1}(R\beta - q) + (R\beta - q)'(R\beta - q)$$

$$(d.2) SSR_r \geq SSR_u$$

$$(d.3) E(SSR_r) = \sigma^2(N-K+J) + (R\beta - q)'(R\beta - q)$$

$$(d.4) \text{Let } \tilde{\sigma}^2 = SSR_r/(N-K+J). \text{ Then, } E(\tilde{\sigma}^2) = \sigma^2 + (R\beta - q)'(R\beta - q)/(N-K+J)$$

The property (d.2) indicates that the restricted regression SSR is *almost surely* greater than the unrestricted SSR. Therefore, the SSR is a nondecreasing, and almost surely a increasing, function of the number of regressors. (d.4) indicates that the restricted regression estimator $\tilde{\sigma}^2$ is an unbiased estimator of σ^2 if the restrictions are correct, and is positively biased (biased upward) if the restrictions are incorrect.

(e) **Distribution Functions.** If $u|X \sim N(0, \sigma^2 I)$, then

(e.1) $\tilde{\beta}$, $\tilde{\lambda}$ and $\tilde{u}$ are distributed as normal.

(e.2) $\tilde{\sigma}^2$ is distributed as a noncentral χ^2 :

$$\frac{(N-K+J)\tilde{\sigma}^2}{\sigma^2} = \frac{SSR_r}{\sigma^2} \sim \chi^2(N-K+J; \delta), \quad \delta = [(R\hat{\beta} - q)' S^{-1} (R\hat{\beta} - q)]^{1/2}$$

(e.3) $\tilde{\beta}$, $\tilde{\lambda}$ and $\tilde{\sigma}^2$ are independently distributed.

The proof of normality in (e.1) is straightforward by using Theorem 2(a) in Statistical Appendix:

$$\tilde{\beta} = \hat{\beta} - VR'S^{-1}(R\hat{\beta} - q) = [I - VR'S^{-1}R](X'X)^{-1}X'u + [\beta + VR'S^{-1}(R\hat{\beta} - q)] \equiv Bu + \xi$$

$$\tilde{\lambda} = \sigma^2 S^{-1}(R\hat{\beta} - q) = \sigma^2 S^{-1}R(X'X)^{-1}X'u + \sigma^2 S^{-1}(R\hat{\beta} - q) = Cu + \epsilon$$

$$\tilde{u} = \hat{u} + XVR'S^{-1}(R\hat{\beta} - q) = \tilde{M}u + \zeta, \quad \zeta = XVR'S^{-1}(R\hat{\beta} - q)$$

The proof of the distribution of $\tilde{\sigma}^2$ uses the last expression of (c.1) and Theorem 3(b). Or, from the first expression in (d.1), we can write

$$\frac{SSR_r}{\sigma^2} = \frac{\hat{u}'\hat{u}}{\sigma^2} + (R\hat{\beta} - q)' S^{-1} (R\hat{\beta} - q) = \frac{(N-K)\hat{\sigma}^2}{\sigma^2} + (R\hat{\beta} - q)' S^{-1} (R\hat{\beta} - q)$$

We have shown earlier that

$$(N-K)\hat{\sigma}^2/\sigma^2 \sim \chi^2(N-K)$$

$$(R\hat{\beta} - q)' S^{-1} (R\hat{\beta} - q) \sim \chi^2(J, \delta)$$

These two χ^2 variates are independent because $\hat{\sigma}^2$ and $\hat{\beta}$ are independent. The desired result then follows from Theorem 6 in Statistical Appendix, the *reproductive property* of the independent noncentral χ^2 random variables

To prove the independence in (e.3), just note that $\tilde{\beta}$, $\tilde{\lambda}$ and $\tilde{u}$ are normal variates and uncorrelated, and hence they are independent. The independence of $\tilde{\sigma}^2$ and $\tilde{\beta}$ (and $\tilde{\lambda}$) follows from Theorem 7 in Statistical Appendix because $\tilde{\sigma}^2$ is a measurable function of $\tilde{u}$.

Restricted LSE by Reparameterization

Restricted coefficients may be estimated from a regression equation modified by inserting the linear restrictions into the regression equation. For example, consider a simple restriction $\beta_1 + \beta_2 = 1$ in a regression equation

$$y_t = \beta_1 x_{t1} + \beta_2 x_{t2} + \beta_3 x_{t3} + \dots + \beta_K x_{tK}$$

Since $\beta_1 = 1 - \beta_2$ under the restriction, the restriction equation can be written as

$$y_t^* \equiv y_t - x_{t1} = \beta_2(x_{t2} - x_{t1}) + \beta_3 x_{t3} + \dots + \beta_K x_{tK}$$

OLSE of this regression equation yields the restricted estimators for $\tilde{\beta}_k$, $k=2,3,\dots,K$, and the restricted estimator of β_1 is computed as $\tilde{\beta}_1 = 1 - \tilde{\beta}_2$.

More generally, partition the restriction equations into

$$R\beta = R_1\beta_1 + R_2\beta_2 = q$$

where R_1 is a $J \times J$ submatrix of R . Rearrange the restriction equations if necessary, such that R_1 is nonsingular. Then, we can solve for β_1 : $\beta_1 = R_1^{-1}(q - R_2\beta_2)$. Substituting this into the regression equation, we have a reparameterized regression model

$$y = X_1\beta_1 + X_2\beta_2 + u = X_1R_1^{-1}(q - R_2\beta_2) + X_2\beta_2 + u = X_1R_1^{-1}q + [X_2 - X_1R_1^{-1}R_2]\beta_2 + u$$

which can be rewritten as

$$y^* \equiv y - X_1R_1^{-1}q = [X_2 - X_1R_1^{-1}R_2]\beta_2 + u = Z\beta_2 + u$$

Once the estimator $\tilde{\beta}_2$ is computed from this equation, we obtain $\tilde{\beta}_1 = R_1^{-1}(q - R_2\tilde{\beta}_2)$.

Example. Suppose that we have data on regressors X_i and regressand y_i for two agents (firms, individuals). Each agent's regression equation is given by

$$y_i = X_i\beta_i + u_i, \quad i=1,2$$

where y_i and u_i are $N_i \times 1$ vectors, X_i is a $N_i \times K$ matrix, and β_i is a $K \times 1$ vector of unknown coefficients. When there are no restrictions on the coefficients, we can estimate β_i from a regression equation

$$y \equiv \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} X_1 & 0 \\ 0 & X_2 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} + \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \equiv X\beta + u$$

The LSE of β_i from this equation is identical to the separate LSE of β_i from each equation.

Suppose that we wish to estimate the coefficients subject K linear restrictions $\beta_1 = \beta_2$, i.e., all K coefficients are equal between the two agents. These restrictions can be written as $R\beta = 0$, where $R = (R_1 : R_2) = (I_K : -I_K)$. The reparameterization is equivalent to running a regression equation of the form

$$y \equiv \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \beta_1 + \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \equiv Z\beta_1 + u$$

Suppose now that we wish to estimate subject to restrictions that only a subset of the K coefficients are equal between the two agents. Partition β_i as

$$\beta_1 = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} \quad \beta_2 = \begin{pmatrix} \gamma_1 \\ \gamma_2 \end{pmatrix}$$

where α_1 and γ_1 are K_1 -dimensional column vectors. We wish to estimate β_1 and β_2 subject to the restrictions $\alpha_1 = \gamma_1$, but allow α_2 and γ_2 to differ. The restricted regression equation can be set up as

$$y \equiv \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} X_{11} & X_{12} & 0 \\ X_{21} & 0 & X_{22} \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \gamma_2 \end{pmatrix} + \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \equiv W\delta + u$$

where X_{11} and X_{12} are, respectively, $N_1 \times K_1$ and $N_1 \times K_2$ submatrices of X_1 , and X_{21} and X_{22} are defined in a similar way.

Statistical Inferences

A major part of statistical inference is the test of hypothesis. A test of hypothesis consists of

- (i) specify the *null hypothesis* (H_0) and the *alternative hypothesis* (H_1)
- (ii) choose a test statistic
- (iii) choose the rejection region for a given level of significance
- (iv) conclude

Each hypothesis specifies the parameter space Θ_0 and Θ_1 , respectively. If Θ_0 is a subset of Θ_1 , then it is called a *nested hypothesis*; otherwise, it is called a *nonnested hypothesis*. A nested null (or maintained) hypothesis $H_0: h(\theta)=0$ against the alternative hypothesis $H_1: h(\theta) \neq 0$ restrict the parameter space¹. To test H_0 against H_1 , we need a *test statistic* that is computable from the sample (i.e., must not involve any unknown parameters), and a decision rule when to reject or not to reject H_0 . The decision rule is expressed in terms of the *rejection region* (or *critical region*), such that H_0 is rejected if the test statistic is in the critical region.

Since the test statistic is a random variable, there is a positive probability that we will reject the true H_0 for any reasonable choice of the critical region. Rejection of a true H_0 is called a *Type I error*, and the probability of Type I error is called the *size of a test*, or the *significance level* of a test, commonly denoted by α . The conventional choice of the significance level is 0.05 or 0.1. We may also fail to reject a false H_0 , which is called a *Type II error*. The probability of rejecting a false H_0 is called the *power of a test*, which depends on the alternative hypothesis.

	Null Hypothesis H_0	
	H_0 is correct	H_0 is false
reject H_0	Type I error	(correct decision)
accept H_0	(correct decision)	Type II error

significance level: $\alpha = \text{prob}(\text{reject a true } H_0) = \text{prob}(\text{type I error})$

Power of test: $\pi = \text{prob}(\text{reject a false } H_0) = 1 - \text{prob}(\text{not reject a false } H_0) = 1 - \text{prob}(\text{type II error})$

Since the significance level is the probability that the test statistic is in the rejection region when the null hypothesis is correct, we need to know the distribution function of the test statistic under H_0 . Similarly, the power of test requires the knowledge about the distribution function of the test statistic under H_1 . Since the distribution function of the test statistic depends on the true value of the parameters, there are many values of the power when the alternative hypothesis is a composite hypothesis.

¹ Review the terminologies such as simple hypothesis, composite hypothesis, joint hypothesis, one-sided (one-tailed) test, and two-sided (two tailed) test.

An *ideal test* (meaning an ideal choice of test statistic and critical region) is the one that has a zero probability of Type I and Type II errors. But such a test does not exist. Typically, choosing a smaller size of a test decreases the power of the test. A test is called the *most powerful test* if it has greater power than any other test of same test size, and the *uniformly most powerful test* if it has greater power than any other test of same test size for all possible values of the parameters. A test is called an *unbiased test* if its power is at least as large as its test size. A test is called a *consistent test* if its power converges to 1 as sample size grows to infinity.

Finite Sample Tests of Nested Linear Hypotheses in a SLM

Linear Regression Model: $y = X\beta + u, \quad u|X \sim N(0, \sigma^2 I)$

Number of observations = N

Summary of previous results

Unrestricted OLS estimator: $\min SSR_u = (y - X\hat{\beta})'(y - X\hat{\beta})$

Properties of Unrestricted OLSE

$$\begin{aligned} \hat{\beta} &= (X'X)^{-1}X'y \sim N(\beta, V), \quad V = \sigma^2(X'X)^{-1} \\ \hat{\sigma}^2 &= SSR_u / (N-K), \quad W_u \equiv (N-K)\hat{\sigma}^2 / \sigma^2 \sim \chi^2(N-K) \\ \hat{\beta} &\text{ is independent of } \hat{\sigma}^2 \end{aligned}$$

$$\begin{aligned} R\hat{\beta} &\sim N(R\beta, S), \quad S \equiv RVR' = \sigma^2 R(X'X)^{-1}R' \\ Z &\equiv (R\hat{\beta} - q)'S^{-1}(R\hat{\beta} - q) \sim \chi^2(J; \delta) \\ F &\equiv \frac{Z/J}{W_u/(N-K)} = \frac{1}{J} (R\hat{\beta} - q)' \hat{S}^{-1} (R\hat{\beta} - q) \sim F(J, N-K; \delta) \\ \delta &= [(R\beta - q)'S^{-1}(R\beta - q)]^{1/2} \end{aligned}$$

Restricted OLS estimator: $\min SSR_r = (y - X\hat{\beta})'(y - X\hat{\beta}) \text{ subject to } R\hat{\beta} = q$

$R(J \times K)$ and $q(J \times 1)$ are known matrices, $rank(R) = J < K$.

Properties of Restricted OLSE

$$\begin{aligned} \tilde{\beta} &= \hat{\beta} - VR'S^{-1}(R\hat{\beta} - q) \sim N(\beta - VR'S^{-1}(R\beta - q), \quad V - VR'S^{-1}RV) \\ \tilde{\lambda} &= \sigma^2 S^{-1}(R\hat{\beta} - q) \sim N(\sigma^2 S^{-1}(R\beta - q), \quad \sigma^4 S^{-1}) \\ \tilde{\sigma}^2 &= \frac{SSR_r}{N-(K-J)}, \quad W_r \equiv \frac{(N-K+J)\tilde{\sigma}^2}{\sigma^2} = \frac{SSR_r}{\sigma^2} \sim \chi^2(N-K+J; \delta) \\ \tilde{\beta}, \tilde{\lambda}, \tilde{\sigma}^2 &\text{ are mutually independent.} \end{aligned}$$

$$SSR_r = SSR_u + \sigma^2(R\hat{\beta} - q)'S^{-1}(R\hat{\beta} - q)$$

Nested Linear Hypotheses:

$$H_0: R\beta = q, \quad H_1: R\beta \neq q \text{ (or } H_0 \text{ is false)}$$

where R is a $J \times K$ matrix of known constants and q is a $J \times 1$ vector of known constants. The restriction matrix R has a full row rank that is less than K , $\text{rank}(R) = J < K$.

An intuition suggests that we may compare the unrestricted estimator $\hat{\beta}$ and restricted estimator $\tilde{\beta}$ to verify whether the restriction under H_0 is acceptable or not. These two estimators will be different with probability 1 due to the random variation of the sample. But, what do we expect about the magnitude of the difference? A big difference or small difference? It is easy to imagine that the difference will be relatively small (relative to their variances) if the restriction under H_0 is correct and it is not binding in the minimization of SSR . On the other hand, the difference is expected to be relatively large if the null hypothesis is incorrect and the null restriction is a binding constraint in the optimization. Then, it is reasonable to consider the difference $\hat{\beta} - \tilde{\beta}$ as a test statistic and construct a rejection region. Of course, we have to decide how to measure the difference because it is a K -dimensional vector. As we will discuss in more detail later, the Durbin-Wu-Hausman test can be interpreted as a test based on this type of reasoning, though it is used for different type of hypotheses.

There is no reason why we should limit the test only to the comparison of the estimators themselves. It is possible to compare the estimated values of a certain function of the parameters. Four tests presented below compare different functions of estimators. We will call them the *likelihood ratio (LR) type test*, *Wald type test*, *Lagrange Multiplier (LM) type test* and *efficient score type test*. The reason for the word 'type' is because the ideas of the finite sample tests discussed below are based on the ideas of the classical test procedures, but they are not the true LR, Wald, LM and Score tests which are asymptotic tests and use the asymptotic χ^2 distribution of the test statistics.

(1) Likelihood Ratio (LR) Type Test

Since the least squares method tries to minimize the SSR , we may compare the values of the objective function (SSR). We know SSR_r is greater than SSR_u . Their difference is expected to be large if the restrictions under H_0 is not true and hence they are binding. Thus, it is intuitively reasonable to use the difference $SSR_r - SSR_u$ as the test statistic and we reject the null hypothesis if $SSR_r - SSR_u$ is large.

Once the test statistic is chosen, we need to determine the decision rule of when to reject H_0 in favor of H_1 . As alluded to already, we will reject H_0 if $SSR_r - SSR_u$ is *sufficiently large*. Otherwise, the test concludes that the data does not provide enough evidence to reject H_0 . How big a difference is sufficiently large enough to reject the null hypothesis? This requires a choice of the critical (rejection) region of the test. Since we reject the null hypothesis when $SSR_r - SSR_u$ is large, the proper critical region is the set of the sample, $R_c = \{SSR_r - SSR_u > c\}$, where c is a pre-chosen critical value. The choice of the critical value depends on how

much of Type I error we are willing to tolerate. A choice of large critical value will reject the null hypothesis less frequently and we are more likely to commit the Type I error. The choice of the critical value is governed by the choice of *significance level (size of the test, or test size)*. That is, for a pre-chosen significance level α , the critical value is determined by

$$P(SSR_r - SSR_u > c \mid H_0) = \alpha$$

which requires the knowledge of the distribution function of the test statistic under the null hypothesis.

We have shown

$$Z \equiv \frac{SSR_r - SSR_u}{\sigma^2} = (R\hat{\beta} - q)' S^{-1} (R\hat{\beta} - q) \sim \chi^2(J, \delta)$$

Recall that $S = \sigma^2 R(X'X)^{-1}R'$ involves the unknown parameter σ^2 . Hence, this is not a statistic (i.e., not computable from the sample). We may just replace σ^2 with its estimator $\hat{\sigma}^2$. But, then, Z does not have the χ^2 distribution. To see this, note that the replacement of σ^2 with its estimator $\hat{\sigma}^2$ is equivalent to taking a ratio of two independent χ^2 random variables, Z and W_u :

$$F \equiv \frac{Z/J}{W_u/(N-K)} = \frac{1}{J} (R\hat{\beta} - q)' \hat{S}^{-1} (R\hat{\beta} - q) \sim F(J, N-K; \delta)$$

which can also be expressed as

$$F \equiv \frac{Z/J}{W_u/(N-K)} = \frac{(SSR_r - SSR_u)/J}{SSR_u/(N-K)} = \frac{N-K}{J} \frac{SSR_r - SSR_u}{SSR_u} \sim F(J, N-K; \delta)$$

This is a statistic that has a central F distribution under H_0 and a noncentral F distribution under H_1 . The null hypothesis is rejected if F is greater than the critical value F_α of significance level α : $P\{F > F_\alpha \mid H_0\} = \alpha$. To compute this test statistic, we need both unrestricted estimators for SSR_u and restricted estimators for SSR_r .

(2) Wald type Test

We know that restricted estimator satisfies the restriction under H_0 , i.e., $R\tilde{\beta} = q$. The Wald type test asks whether the unrestricted estimator also satisfies the restriction equation reasonably well. If the null hypothesis is correct so that the restricted and unrestricted estimators are close enough, we would expect that $R\hat{\beta}$ is close to its hypothesized value q . If they are not close enough we will reject the null hypothesis. We may also interpret this test as a comparison of $R\hat{\beta} - q$ with $R\tilde{\beta} - q$, where the latter is a zero vector.

Unlike the scalar values of SSR's in the LR type test, $R\hat{\beta}$ is a $J \times 1$ vector in general. Therefore, we need to choose a measure of the distance of $R\hat{\beta}$ from q . One may just consider the squared Euclidean norm, $\|R\hat{\beta} - q\|^2 = (R\hat{\beta} - q)'(R\hat{\beta} - q)$. However, this does not take into account the differences in the variances of each element. We need to standardize $R\hat{\beta} - q$ in a certain way. Let $P'P = S^{-1}$. Then $P(R\hat{\beta} - q) \sim N(P(R\hat{\beta} - q), I)$. We now take the Euclidian norm of this standardized vector

$$Z = \|P(R\hat{\beta} - q)\|^2 = (R\hat{\beta} - q)' S^{-1} (R\hat{\beta} - q) \sim \chi^2(J, \delta)$$

As in the previous case of the LR type test, we take the ratio of independent χ^2 random variables to eliminate the unknown σ^2 in S . The test statistic is then exactly same as the LR type test statistic

$$F = \frac{1}{J} (R\hat{\beta} - q)' \hat{S}^{-1} (R\hat{\beta} - q) \sim F(J, N-K; \delta)$$

Under the null hypothesis this F -statistic has a central $F(J, N-K)$. The null hypothesis is rejected if F is greater than the critical value F_α of significance level α : $P\{F > F_\alpha | H_0\} = \alpha$. To compute this test statistic, we need only unrestricted estimators.

(3) Lagrange Multiplier (LM) type Test

As the name indicates, this test is based on the estimator $\tilde{\lambda}$ of the Lagrange multiplier λ . If the restriction of H_0 is true, it will not be a binding constraint and hence, the restricted estimator $\tilde{\lambda}$ (estimator of the shadow price of the restrictions) will be close to zero. Thus, we reject the null hypothesis if $\tilde{\lambda}$ is sufficiently different from a zero vector¹. As in the Wald type test, we need to find a measure of the distance between $\tilde{\lambda}$ and a zero vector because $\tilde{\lambda}$ is a vector in general.

Following the same procedure as in the Wald type test, we note

$$Z_r \equiv \tilde{\lambda}' [\sigma^4 S^{-1}]^{-1} \tilde{\lambda} = \sigma^{-4} \tilde{\lambda}' S \tilde{\lambda} = \sigma^{-2} \tilde{\lambda}' [R(X'X)^{-1} R'] \tilde{\lambda} \sim \chi^2(J, \delta)$$

and we take the ratio of independent χ^2 random variables to eliminate the unknown σ^2

$$F_r = \frac{Z_r/J}{W_r/(N-K+J)} = \frac{1}{J} \tilde{\sigma}^{-2} \tilde{\lambda}' [R(X'X)^{-1} R'] \tilde{\lambda} \sim F(J, N-K+J) \text{ under } H_0$$

Note that this test statistic requires only restricted estimators and its degrees of freedom are different from those of the LR and Wald type tests. Under the null hypothesis, this has a central F distribution because both the numerator and denominator terms are distributed as a central χ^2 under H_0 . However, under the alternative hypothesis ($R\beta \neq q$), both Z_r and W_r are noncentral χ^2 random variables with the same noncentrality parameter δ . Therefore, F_r has a *doubly-noncentral F distribution* under H_1 . This will affect the power of the test. The critical value for the rejection region is still found from the null central F distribution.

We have shown earlier that the LR type test and the Wald type test are identical. To see the relationship of the LM type test with other tests, note the solution $\tilde{\lambda} = \sigma^2 S^{-1} (R\hat{\beta} - q)$. Since S^{-1} has a full column rank, $\tilde{\lambda} \approx 0$ is equivalent to $R\hat{\beta} - q \approx 0$, which is the Wald type test. Therefore, both tests are checking on the same quantity. This becomes clearer if $\tilde{\lambda} = \sigma^2 S^{-1} (R\hat{\beta} - q)$ is substituted into Z_r equation. The resulting equation

¹ Note that this test may be interpreted as a comparison of restricted and unrestricted estimators of λ because the unrestricted estimator $\hat{\lambda}$ is identically zero.

is $Z_r = (R\hat{\beta} - q)'S^{-1}(R\hat{\beta} - q)$, which is the same as Z . Thus, the numerator terms of F and F_r are the same. The difference is the denominator term. If we use the unrestricted estimator $\hat{\sigma}^2$ instead of the restricted estimator $\tilde{\sigma}^2$, that is, if we use $W_u/(N-K)$ instead of $W_r/(N-K+J)$ in the ratio F_r , then we have a test statistic that is the same as the Wald type test.

(4) (Efficient) Score Type Test

The unrestricted estimator is the solution of the first order condition

$$\left. \frac{\partial SSR}{\partial \beta} \right|_{\beta=\hat{\beta}} = -2X'y + 2X'X\hat{\beta} = 0 \Rightarrow g(\hat{\beta}) \equiv -X'y + X'X\hat{\beta} = 0$$

If the restrictions are correct, then the restricted estimator must satisfy this equation closely so that

$$g(\tilde{\beta}) = -X'y + X'X\tilde{\beta} \approx 0$$

The score type test thus asks whether the restricted estimator satisfies the first order condition of the unrestricted least squares estimation. The test rejects the null hypothesis if $g(\tilde{\beta})$ is far from a zero vector. This test can also be considered as a comparison of restricted gradient $g(\tilde{\beta})$ and unrestricted gradient $g(\hat{\beta})$ which is of course zero.

The score type test is equivalent to the LM type test. Consider the first order condition of the restricted least squares

$$-X'y + X'X\tilde{\beta} + R'\tilde{\lambda} \equiv g(\tilde{\beta}) + R'\tilde{\lambda} = 0$$

Since R' has a full column rank by our specification of the null hypothesis, $\tilde{\lambda} = 0$ if and only if $g(\tilde{\beta}) = 0$. Therefore, the LM type test of asking whether $\tilde{\lambda}$ is close to zero or not is equivalent to asking whether $g(\tilde{\beta})$ is close to zero or not. Substitution of $R'\tilde{\lambda} = -g(\tilde{\beta})$ into the Z_r gives

$$Z_r = \sigma^{-2}\tilde{\lambda}'[R(X'X)^{-1}R']\tilde{\lambda} = \sigma^{-2}(R'\tilde{\lambda})'(X'X)^{-1}(R'\tilde{\lambda}) = \sigma^{-2}g(\tilde{\beta})'(X'X)^{-1}g(\tilde{\beta})$$

and

$$F_r = \frac{1}{J}\tilde{\sigma}^{-2}g(\tilde{\beta})'(X'X)^{-1}g(\tilde{\beta}) \sim F(J, N-K+J) \text{ under } H_0$$

Case of Single Restriction: t test

If $J=1$ and $R\hat{\beta}$ is a scalar

$$t^* = \frac{(R\hat{\beta} - R\beta)/sd(R\hat{\beta})}{\sqrt{W/(N-K)}} = \frac{(R\hat{\beta} - R\beta)/sd(R\hat{\beta})}{\sqrt{\hat{\sigma}^2/\sigma^2}} = \frac{(R\hat{\beta} - R\beta)}{est. sd(R\hat{\beta})} \sim t(N-K)$$

$$t = \frac{(R\hat{\beta} - q)}{est. sd(R\hat{\beta})} \sim t(N-K, R\beta - q)$$

H_0 is rejected when $|t| > t_\alpha$ where the critical value is determined by $P(|t| > t_\alpha | H_0) = \alpha$.

Homework #2

[1] Explain the intuitive ideas of the LR-type test, Wald-type test, LM-type test and Score-type test. Do not hand in your answers to this question. But, I ask this question very often in the qualifying exam.

[2] Consider a linear regression model $y = X\beta + u$, where the first regressor is a constant ($X_{it} = 1$) and $u|X \sim N(0, \sigma^2 I)$. Present the test statistic, how to compute it, its distribution under H_0 and H_1 , and rejection region. You may want to specify the restriction equation under H_0 in the form of $R\beta = q$ by specifying R and q . When the number of restriction under the null hypothesis is 1, present both t -test and F -test.

(a) Test of an individual coefficient

$$(i) H_0: \beta_k = c, \quad H_1: \beta_k > c \quad \dots \text{One sided test}$$

$$(ii) H_0: \beta_k = c, \quad H_1: \beta_k \neq c \quad \dots \text{Two sided test}$$

where c is a known constant.

(b) Test of a linear function of coefficients

$$H_0: a\beta_i + b\beta_j = c, \quad H_1: a\beta_i + b\beta_j \neq c$$

where a , b and c are known constants.

(c) Test of the significance of all *slope* coefficients

$$H_0: \beta_2 = \beta_3 = \dots = \beta_K = 0, \quad H_1: H_0 \text{ is false}$$

In addition to the questions posed at the beginning, show the relationship between the F -test statistic and R^2 .

(d) Test of the stability of entire coefficients between two sub-samples

$$\text{Model: } y_t = X_t \beta_t + u_t, \quad t = 1, 2, \dots, N, \quad u_t \sim i.i.d.N(0, \sigma^2)$$

The subscript t for β indicates that the regression coefficients may change over time.

$$H_0: \beta_t = \beta \text{ for all } t \dots \text{Coefficients are stable}$$

$$H_1: \beta_t = \begin{cases} \beta_1 & \text{if } t \leq N_1 \\ \beta_2 & \text{if } t > N_1 \end{cases}$$

where $N_1 > K$ and $N_2 = N - N_1 > K$, that is, the sample size in each subperiod is greater than the number of regressors. Partition the observation matrix of y and regressors as

$$y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}, \quad X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$$

where $y_1: N_1 \times 1$, $y_2: N_2 \times 1$, $X_1: N_1 \times K$, $X_2: N_2 \times K$, and $\text{rank}(X_1) = \text{rank}(X_2) = K$. You may need to add more assumptions to answer this question

(e) Test of the stability of a subset of coefficients between two sub-samples

Model: $y_t = X_t \beta + W_t \gamma + u_t$, where $X_t: 1 \times K$, $W_t: 1 \times L$, $t = 1, 2, \dots, N$

Hypotheses:

$H_0: \beta_t = \beta, \gamma_t = \gamma \dots$ all coefficients are stable

$H_1: \beta_t = \beta, \gamma_t = \gamma_1 \text{ for } t \leq N_1 \text{ and } \beta_t = \beta, \gamma_t = \gamma_2 \text{ for } t > N_1$

where $N_1 > K + L$ and $N_2 = N - N_1 > K + L$.

(f) Test of the stability of entire coefficients between two sub-samples *with insufficient degrees of freedom*

The model and hypotheses are the same as those in (d), except that $N_2 < K$, a smaller number of observations than the number of regressors in the second subsample. (refer to Assumption 1 for the implication of this).

We maintain the assumption $\text{rank}(X_1) = K$. Consider two cases:

(i) Case 1: $\text{rank}(X_2) = N_2 < K$

(ii) Case 2: $\text{rank}(X_2) < N_2 < K$

(g) Test of the stability of entire coefficients among multiple sub-samples (*with insufficient degrees of freedom*)

$y_j = X_j \beta_j + u_j, j = 1, 2, \dots, m \dots m$ subsamples

$y_j: N_j \times 1, X_j: N_j \times K, \beta_j: K \times 1, u_j: N_j \times 1, \sum_{j=1}^m N_j = N$

$r_j = \text{rank}(X_j) \leq \min[N_j, K]$

$H_0: \beta_1 = \beta_2 = \dots = \beta_m$

$H_1: H_0 \text{ is false (i.e., } \beta_i \neq \beta_j \text{ for at least one pair of } i \text{ and } j)$

(i) Case 1. $r_j = K \leq N_j$ for all j

(ii) Case 2: $r_j = N_j < K$ for some j

(iii) Case 3: $r_j < N_j < K$ for some j

(h) Consider the case in (d). Let $\hat{\beta}_1$ be the OLS estimator from the first subsample, and let $\hat{u}_1 = y_1 - X_1 \hat{\beta}_1$ be the residual vector. Let $e_2 = y_2 - \hat{y}_2$ be the prediction error, where $\hat{y}_2 = X_2 \hat{\beta}_1$ is the BLUP. We have shown earlier that $E(e_2) = 0$ and $\text{cov}(e_2) = \sigma^2 I_{N_2} + \sigma^2 X_2 (X_1' X_1)^{-1} X_2'$. It is easy to see that e_2 is distributed as a normal.

Therefore, its quadratic form

$$\chi_e^2 = e_2' [\sigma^2 I_{N_2} + \sigma^2 X_2 (X_1' X_1)^{-1} X_2']^{-1} e_2 \sim \chi^2(N_2, \delta)$$

is distributed as a noncentral χ^2 with the noncentrality parameter

$$\delta^2 = [X_2(\beta_2 - \beta_1)]' [\sigma^2 I_{N_2} + \sigma^2 X_2 (X_1' X_1)^{-1} X_2']^{-1} [X_2(\beta_2 - \beta_1)]$$

We know

$$\chi_u^2 = \hat{u}_1' \hat{u}_1 / \sigma^2 = (N_1 - K) \hat{\sigma}^2 / \sigma^2 \sim \chi^2(N_1 - K)$$

is distributed as a central χ^2 , and it is independent of $\hat{\beta}_1$.

- (i) χ_u^2 and χ_e^2 are independent. Why?
 - (ii) Construct a F -test of the null hypothesis $\beta_1 = \beta_2$ by using these results. You need to specify the test statistic, its null and alternative distributions, and the rejection region.
 - (iii) Suppose $\text{rank}(X_2) = N_2 < K$. Compare your test statistic for this problem with the test statistic in Case 1 of (f)
-

Nonlinear Hypothesis and Delta Method

Suppose that the null hypothesis is a continuous nonlinear function of β , $H_0: h(\beta)=0$. For example, $h(\beta)=\beta_1^2+\beta_2-1=0$. One way to conduct the test is by using the delta method and asymptotic normal or t distribution.

Delta Method

Consider a linear regression model $y=X\beta+u$, for which the OLS estimator $\hat{\beta}_N$ from a sample of size N is a consistent estimator of β and its asymptotic distribution is normal by the central limit holds:

$$\sqrt{N}(\hat{\beta}_N - \beta) \xrightarrow{d} N(0, \Sigma_\beta)$$

For notational simplicity, the subscript N will be dropped from $\hat{\beta}_N$. Let $h(\beta)$ is a continuous *scalar* function of β . Then, $h(\hat{\beta})$ is a consistent estimator of $h(\beta)$ and its asymptotic distribution is derived by the first order Taylor expansion as follows:

$$h(\hat{\beta}) \approx h(\beta) + h'(\beta)(\hat{\beta} - \beta), \quad h'(\beta) = \frac{\partial h(\beta)}{\partial \beta}$$

which gives the asymptotic variance

$$\sqrt{N}(h(\hat{\beta}) - h(\beta)) \xrightarrow{d} N(0, \sigma_h^2), \quad \sigma_h^2 = h'(\beta) \Sigma_\beta h(\beta)$$

The null hypothesis $h(\beta)=0$ is then tested by using $h(\hat{\beta})/\hat{\sigma}_h$ as the test statistic. Either standard normal or t distribution is used for the test. This can be extended to the case where $h(\beta)$ is a vector function, which will be discussed more in the section of the maximum likelihood estimation.

Bootstrap Method

When the normality assumption is invalid, or the distribution of error term is unknown, or the hypothesis is nonlinear, we typically rely on the asymptotic justification for the distribution of test statistic. This asymptotic approximation can be pretty good, but it can also be pretty bad for a finite sample. An alternative method is the bootstrap method. The bootstrap method is more commonly used for computing the standard errors and/or confidence intervals of the estimators $\hat{\beta}_k$ or $h(\hat{\beta})$. But it can also be used as a tool for the test.

There are two types of bootstrapping: parametric and nonparametric. We will consider an example of finding standard error and confidence interval in a linear regression model $y_i = X_i\beta + u_i$.

Step 1: Compute the OLS estimate $\hat{\beta}$ and $h(\hat{\beta})$, and compute residuals $\hat{u}_i = y_i - X_i\hat{\beta}$.

Step 2: Construct a new residual vector $\tilde{u}$ by randomly selecting elements from $\hat{u}$ with replacement, and new *bootstrap sample* for y : $\tilde{y}_i = X_i\hat{\beta} + \tilde{u}_i$.

Step 3: Compute new estimate $\tilde{\beta}$ and $h(\tilde{\beta})$.

Step 4: Repeat steps 2 and 3, collecting $\tilde{\beta}$ and $h(\tilde{\beta})$.

Step 5: Compute the empirical standard errors and confidence interval from the collected $\tilde{\beta}$ and $h(\tilde{\beta})$.

Note: To construct new residual vector $\tilde{u}$

Nonparametric bootstrap - resample from $\hat{\mathbf{u}}_t$ with replacement

Parametric bootstrap - if $\mathbf{u}_t$ is assumed to be i.i.d. $N(\mathbf{0}, \sigma^2)$, then generate normal random numbers with zero mean and variance $\hat{\sigma}^2$.

Tests of Nonnested Hypothesis

If the model under the null hypothesis is a special case of a model under the alternative hypothesis, then the hypothesis is *nested*. The *nonnested* models are the case when one model can not be derived from the other model by imposing certain restrictions. Examples of nonnested linear and nonlinear regression models are:

Linear regression model (LRM)

$$H_0: y = X\beta + u_0, \quad E(u_0) = 0, \quad \text{cov}(u_0) = \sigma_0^2 I \quad (1a)$$

$$H_1: y = Z\gamma + u_1, \quad E(u_1) = 0, \quad \text{cov}(u_1) = \sigma_1^2 I \quad (1b)$$

Nonlinear regression model (NRM)

$$H_0: y = f(X, \beta) + u_0, \quad E(u_0) = 0, \quad \text{cov}(u_0) = \sigma_0^2 I \quad (2a)$$

$$H_1: y = g(Z, \gamma) + u_1, \quad E(u_1) = 0, \quad \text{cov}(u_1) = \sigma_1^2 I \quad (2b)$$

lognormal against exponential

$$H_0: f(y_t | \theta_0) = \frac{1}{\sqrt{2\pi\sigma^2} y_t} \exp\left(-\frac{(\ln y_t - \mu)^2}{2\sigma^2}\right), \quad y_t > 0, \quad \sigma^2 > 0$$

$$H_1: g(y_t | \gamma) = \frac{1}{\gamma} \exp\left(-\frac{y_t}{\gamma}\right), \quad y_t > 0, \quad \gamma > 0$$

Different functional forms: linear against log-linear regression model

$$H_0: y = X\beta + u_0, \quad E(u_0) = 0, \quad \text{cov}(u_0) = \sigma_0^2 I$$

$$H_1: \ln(y) = Z\gamma + u_1, \quad E(u_1) = 0, \quad \text{cov}(u_1) = \sigma_1^2 I$$

Probit against logit models (usually $X_t = Z_t$)

$$H_0: P(y_t = 1) = \Phi(X_t \beta) = \int_{-\infty}^{X_t \beta} \frac{1}{2\pi} e^{-z^2/2} dz$$

$$H_1: P(y_t = 1) = L(Z_t \gamma) = \frac{e^{Z_t \gamma}}{1 + e^{Z_t \gamma}}$$

Autoregressive vs Moving Average Errors

$$y_t = X_t \beta + u_t$$

$$H_0: u_t = \sum_{s=1}^p \rho_s u_{t-s} + \epsilon_t, \quad \epsilon_t \sim NID(0, \sigma^2)$$

$$H_1: u_t = \epsilon_t - \sum_{s=1}^p \tau_s \epsilon_{t-s} + \epsilon_t, \quad \epsilon_t \sim NID(0, \sigma^2)$$

An example of the application of nonnested tests of linear models is the test of Keynesian and new classical macroeconomic models. Keynesian model assumes that money growth rates have real effects due to price stickiness, market imperfection, etc. The new classical macro theory on the other hand is based on

the optimizing behavior of economic agents who adjust their decisions according to their expected money growth rates. The expected growth rates thus have no real effect; only unexpected monetary shocks can have real effects. This difference between these two models can thus be expressed in the form of unemployment equation

$$\text{Keynesian model:} \quad u_t = W_t \beta_1 + \beta_{20} m_t + \beta_{21} m_{t-1} + \beta_{22} m_{t-2} + \epsilon_t$$

$$\text{New Classical model:} \quad u_t = W_t \gamma_1 + \gamma_{20} um_t + \gamma_{21} um_{t-1} + \gamma_{22} um_{t-2} + \xi_t$$

where u_t is the rate of unemployment, W_t is a set of variables common to both models (such as time trend, dummy variables), m_t is the money growth rate and um_t is the unexpected rate of money growth.

We will consider the test of non-nested hypothesis of linear regression model

$$H_0: y = X\beta + u_0, \quad E(u_0) = 0, \quad \text{cov}(u_0) = \sigma_0^2 I$$

$$H_1: y = Z\gamma + u_1, \quad E(u_1) = 0, \quad \text{cov}(u_1) = \sigma_1^2 I$$

Two methods of test will be discussed below: Artificial Nesting Method and Encompassing Method.

[1] *Artificial Nesting Method*

This method embeds both models in a more general model, called the *artificial compound model* or the *comprehensive model*, and test each model against the general one. This is suggested in Atkinson. In the linear models, the artificial compound model may be specified as a weighted average of the means under each hypothesis

$$y = (1-\phi)X\beta + \phi Z\gamma + u = X[(1-\phi)\beta] + Z(\phi\gamma) + u = X\beta_c + Z\gamma_c + u \quad (3)$$

where ϕ is a scalar *mixing* parameter. We assume that the columns of X are not a linear combination of the columns of Z , and vice versa. In particular, this assumption excludes the case of common variables in X and Z . We will discuss the case of common variables later.

In this compound model, $\phi=0$ means the model H_0 , and $\phi=1$ means the model H_1 . Thus the test of H_0 against H_1 is equivalent to testing $\phi=0$. However, ϕ and γ are not separately identifiable in (3) and they cannot be estimated separately.

One solution to this problem is to test $\gamma_c=0$ from the last expression by using an F test for the test of H_0 against H_1 . One may also test $\beta_c=0$ which is an encompassing test of H_1 as the null hypothesis and H_0 as an alternative hypothesis. If $\beta_c=0$, then model H_1 is proper, and if $\gamma_c=0$, then model H_0 is proper. The test may reject both $\beta_c=0$ and $\gamma_c=0$, which is an indication that both models are not properly specified. The test may not reject either one. In such a case we conclude that the data does not provide enough information to distinguish between the two models. A drawback of this artificial nesting equation is a possibility of multicollinearity (high correlation among regressors), which causes imprecise estimates of coefficients (large estimated standard errors of the coefficients).

An alternative solution to the identifiability problem is to test $\phi=0$ conditional on an unbiased and/or consistent estimate of γ . Given an estimator $\hat{\gamma}$, we test $\phi=0$ from

$$y = X\beta_c + \phi(Z\hat{\gamma}) + u = X\beta_c + \phi\hat{y}_z + u \quad (4)$$

by using the usual t -test. The test of $\phi=0$ in (4) is called J test because it involves joint estimation of β_c and ϕ . One can use any consistent estimator of γ as long as $Z\hat{\gamma}$ is asymptotically uncorrelated with the errors u_0

under the null hypothesis. Two estimators of γ are commonly used in the literature.

The **J test** of Davidson and MacKinnon (1981) uses the OLS estimator $\hat{\gamma}$ under H_1 :

$$\hat{\gamma}_z = Z\hat{\gamma} = Z(Z'Z)^{-1}Z'y \equiv P_z y \quad (5)$$

This estimator is an unbiased estimator of $Z\gamma$ under H_1 , i.e., $E(Z\hat{\gamma} | H_1) = Z\gamma$, and it is uncorrelated with u_0 because of the assumption of exogeneity of X and Z .

The **JA test** of Fisher and McAleer (1981) uses an estimator of the expectation of (5) under H_0 , which is given by

$$E(Z\hat{\gamma} | H_0) = E[P_z(X\beta + u_0)] = P_z X\beta \quad (6)$$

An estimator of (6) that is unbiased under H_0 is obtained by replacing β with null estimator $\hat{\beta} = (X'X)^{-1}X'y$, so that

$$\tilde{\gamma}_z = Z\tilde{\gamma} = P_z X\hat{\beta} = P_z P_x y \quad (7)$$

where

$$\tilde{\gamma} = (Z'Z)^{-1}Z'X\hat{\beta} = (Z'Z)^{-1}Z'X(X'X)^{-1}X'y$$

is the OLSE of $X\hat{\beta}$ on Z . That is, $\tilde{\gamma}$ is the solution to the minimization of the sum of squared deviations of $\hat{\gamma}_x = X\hat{\beta}$ from $Z\gamma$

$$\min (\hat{\gamma}_x - Z\gamma)'(\hat{\gamma}_x - Z\gamma)$$

Note also that the **JA test** cannot be used if Z and X are orthogonal ($Z'X=0$) because $\tilde{\gamma}_z=0$ in that case.

One may go one step further by considering the expected value of $\tilde{\gamma}_z$ under H_1 :

$$E(\tilde{\gamma}_z | H_1) = E[P_z P_x (Z\gamma + u_1)] = P_z P_x Z\gamma$$

and use its unbiased estimator

$$\check{\gamma}_z = P_z P_x P_z y$$

Another approach to the identifiability problem is to treat γ and β in (3) symmetrically. That is, in addition to replacing γ by its estimate $\hat{\gamma}$ under H_1 , we also substitute β by its estimate $\hat{\beta}$ under H_0 . Then, we can write (3) as

$$y = (1-\phi)X\hat{\beta} + \phi Z\hat{\gamma} + v, \quad \text{or} \quad y - X\hat{\beta} = \phi(Z\hat{\gamma} - X\hat{\beta}) + v \quad (8)$$

and test $\phi=0$ by the t test. Davidson and MacKinnon call this the **C test** because it is a test conditional on the estimate of β . They show that the test size (significance level) of this test is smaller than the nominal size. That is, it is a conservative test in the sense that it rejects the correct null hypothesis less often than the chosen significance level, say, 5%.

Models with Common Regressors

One of the conditions we imposed in the discussion above is that the columns of X is not a linear combination of the columns of Z , and vice versa. If they have some common variables, this condition is not satisfied and the test statistic must be modified. Let $X=(W, X_2)$ and $Z=(W, Z_2)$ where W is the variables common to both null and alternative hypothesis:

$$H_0: y = W\beta_1 + X_2\beta_2 + u_0 \quad (9a)$$

$$H_1: y = W\gamma_1 + Z_2\gamma_2 + u_1 \quad (9b)$$

In this case, the artificial compound model is specified as

$$y = (1-\phi)(W\beta_1 + X_2\beta_2) + \phi(W\gamma_1 + Z_2\gamma_2) + u$$

which can be written as either in the form of encompassing test

$$y = W[(1-\phi)\beta + \phi\gamma_1] + X_2[(1-\phi)\beta_2] + Z_2(\phi\gamma_2) = W\alpha_1 + X_2\alpha_2 + Z_2\alpha_3 + u \quad (10a)$$

or in the form of J test

$$y = W[(1-\phi)\beta + \phi\hat{\gamma}_1] + X_2[(1-\phi)\beta_2] + \phi(Z_2\hat{\gamma}_2) = W\alpha_1 + X_2\alpha_2 + \phi(Z_2\hat{\gamma}_2) + u \quad (10b)$$

where $\hat{\gamma}_i$ is the OLSE of γ_i under H_1 . The Atkinson test tests $\alpha_3=0$ in (10a) with an F test, and the J test tests $\phi=0$ in (10b) with a t test.

The estimator $\hat{\gamma}_2$ is given by

$$\hat{\gamma}_2 = (Z_2' M_w Z_2)^{-1} Z_2' M_w y \quad M_w = I - W(W'W)^{-1}W'$$

This estimator is unbiased estimator of γ_2 under H_1 , but it is biased under H_0 with a mean

$$E(\hat{\gamma}_2|H_0) = (Z_2' M_w Z_2)^{-1} Z_2' M_w X_2 \beta_2$$

An unbiased estimator of this under H_0 is

$$\tilde{\gamma}_2 = (Z_2' M_w Z_2)^{-1} Z_2' M_w X_2 \hat{\beta}_2 = (Z_2' M_w Z_2)^{-1} Z_2' M_w X_2 (X_2' M_w X_2)^{-1} X_2' M_w y$$

where $\hat{\beta}_2$ is the OLSE of β_2 in (9a). Note that this estimator can be computed easily by the OLSE of the coefficient of variable Z_2 in an equation with $X_2\hat{\beta}_2$ as the dependent variable and (W, Z_2) as the regressors.

The JA test of H_0 against H_1 in (9) thus tests $\phi=0$ in a regression equation

$$y = W\alpha_1 + X_2\alpha_2 + \phi(Z_2\tilde{\gamma}_2) + u \quad (10c)$$

[2]. Encompassing Test

Another approach, called the encompassing principle, to test H_0 against H_1 is to compare the estimators $\hat{\gamma}$ and $\tilde{\gamma}$ and test of their equality, where¹

$$\hat{\gamma} = (Z'Z)^{-1}Z'y \quad (11a)$$

$$\tilde{\gamma} = (Z'Z)^{-1}Z'X(X'X)^{-1}X'y \quad (11b)$$

It was shown earlier that $E(\hat{\gamma}) = E(\tilde{\gamma})$ under H_0 . Under H_1 , however,

$$E(\hat{\gamma}) = \gamma$$

$$E(\tilde{\gamma}) = (Z'Z)^{-1}Z'X(X'X)^{-1}X'Z\gamma$$

and hence, $E(\hat{\gamma}) \neq E(\tilde{\gamma})$ under H_1 . This suggests a test of $E(\hat{\gamma} - \tilde{\gamma}) = 0$ against $E(\hat{\gamma} - \tilde{\gamma}) \neq 0$ as a test of H_0 against H_1 . It is straightforward to show that, under the normality of u_0 , the test of $E(\hat{\gamma} - \tilde{\gamma}) = 0$ is an F test with the test statistic

$$F = \frac{y' M_x Z (Z' M_x Z)^{-1} Z' M_x y}{K_z \hat{\sigma}_0} \quad \approx \quad F(K_z, N-K_x) \quad \text{under } H_0 \quad (12a)$$

where $M_x = I - X(X'X)^{-1}X'$, K_z is the dimension of γ and $\hat{\sigma}_0$ is the estimate of error variance under H_0 .

This encompassing test statistic is equivalent to the F test statistic of $\gamma_c=0$ (or $E(\hat{\gamma}_c)=0$) in the

¹ $\tilde{\gamma}$ is called the value of γ predicted by the model under H_0 .

comprehensive model (3)

$$F = \frac{y' M_x Z (Z' M_x Z)^{-1} Z' M_x y}{K_z \hat{\sigma}_c^2} \quad (12b)$$

except for the estimator of error variance $\hat{\sigma}_c^2$ in the place of $\hat{\sigma}_0^2$.

Exercise

- (1) Verify the encompassing test statistics in (12).
 (2) Formulate the encompassing test in the case of common regressors as specified in (9) by comparing $\hat{\gamma}_2$ and $\tilde{\gamma}_2$.
 (3) Atkinson's comprehensive model approach to the test of nonnested hypotheses is based on an exponential mix of the likelihood functions under each hypothesis

$$L_m(L_0, L_1; \lambda) = K L_0(\theta_0)^{1-\lambda} L_1(\theta_1)^\lambda$$

where $L_i(\theta_i)$ is the likelihood function under hypothesis H_i , $i=0,1$, and K is a constant such that the mixed function L_m is a density function, and λ is the mixing parameter. The mixed log likelihood function is

$$\ell_m(\theta_0, \theta_1, \lambda) = \ln(K) + (1-\lambda)\ell_0(\theta_0) + \lambda\ell_1(\theta_1)$$

The idea of the Aitken test of the nonnested hypothesis H_0 against H_1 is to estimate parameters (including the mixing parameter λ) that maximizes ℓ_m , and then test $\lambda=0$.

Show that the mixed log likelihood function for nonnested linear models in (1) under the normality assumption can be written as

$$\ell_m = -\frac{N}{2}\ln(2\pi) - \frac{N}{2}\ln(\sigma^2) - \frac{(y - (1-\phi)X\beta - \phi Z\gamma)'(y - (1-\phi)X\beta - \phi Z\gamma)}{2\sigma^2}$$

where

$$\sigma^2 = \frac{\sigma_0^2 \sigma_1^2}{(1-\lambda)\sigma_1^2 + \lambda\sigma_0^2}, \quad \phi = \frac{\sigma^2}{\sigma_1^2} \lambda$$

which is the log likelihood function of y with the conditional mean $(1-\phi)X\beta + \phi Z\gamma$ and variance $\sigma^2 I$. Therefore, the maximization of the mixed log-likelihood function is equivalent to a regression function

$$y = (1-\phi)X\beta + \phi Z\gamma + u = X[(1-\phi)\beta] + Z(\phi\gamma) + u$$

which is same as (3) and Aitken's testing $\lambda=0$ is equivalent to testing $\phi=0$.

[4] Test of autoregressive mean function vs serially correlated errors

Consider two models

$$H_0: y_t = W_t\beta_1 + X_t\beta_2 + \beta_3 y_{t-1} + \epsilon_t$$

$$H_1: y_t = W_t\gamma_1 + Z_t\gamma_2 + u_t, \quad u_t = \rho u_{t-1} + \eta_t$$

The alternative model can be considered as a nonlinear model

$$H_1: y_t = W_t\gamma_1 + Z_t\gamma_2 + \rho(y_{t-1} - W_{t-1}\gamma_1 - Z_{t-1}\gamma_2) + \eta_t$$

This case can be considered as a special case of a test of nonnested nonlinear models where specification under H_0 happens to be a linear model. Show that the J test of H_0 against H_1 can be based on the t -test of $\phi=0$ in

$$y_t = W_t\alpha_1 + X_t\alpha_2 + \alpha_3 y_{t-1} + \phi\hat{P}_t + \xi_t$$

where $\hat{P}_t = Z_t\hat{\gamma}_2 - W_{t-1}\hat{\rho}\hat{\gamma}_1 - Z_{t-1}\hat{\rho}\hat{\gamma}_2$, and $\hat{\gamma}_i$ and $\hat{\rho}$ are consistent estimators of parameters of alternative model.

Appendix

Estimation of a subvector of β .

Let X be partitioned as $X=(X_1 : X_2)$, so that

$$y = X_1\beta_1 + X_2\beta_2 + u$$

where X_i is $N \times K_i$. We are interested in the estimator of β_i . Applying the inverse of a partitioned matrix, $(X'X)^{-1}$ can be written as

$$(X'X)^{-1} = \begin{pmatrix} X_1'X_1 & X_1'X_2 \\ X_2'X_1 & X_2'X_2 \end{pmatrix}^{-1} = \begin{pmatrix} (X_1'M_2X_1)^{-1} & - (X_1'X_1)^{-1}X_1'X_2(X_2'M_1X_2)^{-1} \\ - (X_2'M_1X_2)^{-1}X_2'X_1(X_1'X_1)^{-1} & (X_2'M_1X_2)^{-1} \end{pmatrix}$$

where

$$M_i = I - X_i(X_i'X_i)^{-1}X_i', \quad i=1,2$$

$$(X_i'M_jX_j)^{-1} = (X_i'X_i)^{-1} + (X_i'X_i)^{-1}X_i'X_j(X_j'M_iX_j)^{-1}X_j'X_i(X_i'X_i)^{-1}, \quad i \neq j=1,2$$

$$(X_i'X_i)^{-1}X_i'X_j(X_j'M_iX_j)^{-1} = (X_i'M_jX_j)^{-1}X_i'X_j(X_j'X_j)^{-1}$$

The OLS estimator of β is

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = (X'X)^{-1}X'y = \begin{pmatrix} (X_1'M_2X_1)^{-1} & - (X_1'M_2X_1)^{-1}X_1'X_2(X_2'M_1X_2)^{-1} \\ - (X_2'M_1X_2)^{-1}X_2'X_1(X_1'X_1)^{-1} & (X_2'M_1X_2)^{-1} \end{pmatrix} \begin{pmatrix} X_1'y \\ X_2'y \end{pmatrix}$$

so that

$$\hat{\beta}_i = (X_i'M_jX_j)^{-1}X_i'y - (X_i'M_jX_j)^{-1}X_i'X_j(X_j'X_j)^{-1}X_j'y = (X_i'M_jX_j)^{-1}X_i'M_jy \quad i \neq j=1,2$$

Note that M_jy is the residual vector of a regression of y on X_j , and similarly, M_jX_i is the residual matrix of regressions of each column of X_i as a dependent variable against X_j . Therefore, $\hat{\beta}_i$ is obtained from the regression of the residual vector M_jy against the residual matrix M_jX_i :

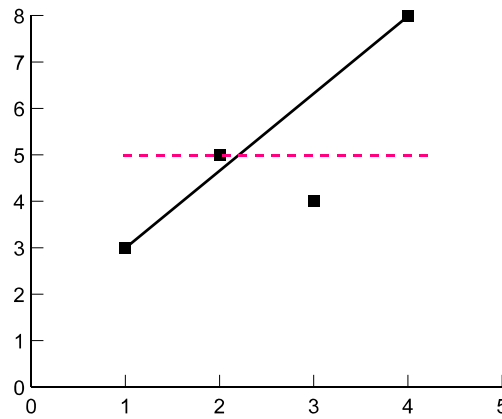
$$M_jy = M_jX_i\beta_i + M_ju$$

which is the equation that arises when M_j is premultiplied to the original regression equation $y = X_i\beta_i + X_j\beta_j + u$, because $M_jX_j=0$.

Supplemental Note

Minimization of the Sum of Residuals

The least squares method minimizes the sum of *squared* residuals. What if we minimize the sum of residuals? It could give a totally unreasonable result because of the cancellation of positive and negative residuals. Consider the following figure of four observations to estimate a simple linear regression model $y = \alpha + \beta x + u$. Compare the two regression lines. The solid line seems to be more reasonable description of the relationship between x and y . However, the sum of residuals is smaller in absolute value (it is zero) for the dashed horizontal line.



The Method of Robust Estimation

An important drawback of the method of least squares is the problem of “outliers”. Since the minimand of the least squares method is the sum of *squared* deviations of the observed values from the estimated line, one observation that lies far from other clusters of the observed values along the regression line can significantly distort the estimated line. Several alternative estimators have been proposed to reduce the effect of the outliers.

(a) L_p Estimators

The least squares estimator can be considered as an estimator that minimizes the sum of the squared *absolute* values of the errors such as $\sum |Y_i - \alpha - \beta X_i|^2$. The L_p estimator minimizes the sum of the absolute values of the errors each raised to the p th power:

$$\min_{\alpha, \beta} \sum_{i=1}^n |Y_i - \alpha - \beta X_i|^p.$$

The value of the power is usually chosen to be between one and two, preferring a lower value when the distribution of the error has fatter tails. The least squares estimator is a special case of the L_p estimator with $p=2$.

If $p=1$, this estimator minimizes the sum of the absolute values of the errors, and it is called the MAD (minimum absolute deviation) estimator, or the LAR (least absolute residual) estimator, or the LAE (least absolute error) estimator.

Lemma. The MAD estimator of the mean is the median.

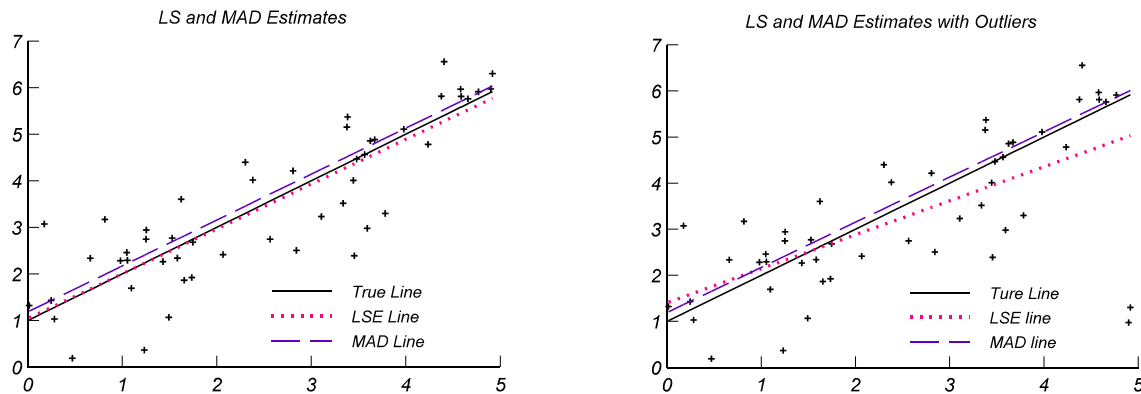
Proof. Differentiation of $\sum_i |X_i - \mu|$ with respect to μ and setting it to zero gives

$$-(\text{number of } X_i \geq \mu) + (\text{number of } X_i \leq \mu) = 0$$

and therefore, the median is the solution to this first order condition. ◻

The MAD estimator of the linear regression function is much more involved and there are many computer algorithms that solve the minimization problem. A straightforward way to find the MAD estimators for α and β in a regression equation $Y_i = \alpha + \beta X_i$ is through an extensive search method. For any given β , the MAD estimator of α is the median of the sample $(Y_i - \beta X_i)$, $i=1,2,\dots,n$, by the lemma above. The median will change with the value of chosen β . Let this median be denoted by $m_d(\beta)$. Then, we can calculate $\sum |Y_i - m_d(\beta) - \beta X_i|$ for a wide range of β , and choose the value of β that gives the smallest sum. The MAD estimator of the regression line ($\hat{Y} = \hat{\alpha} + \hat{\beta}X$) is contrasted with the LSE on the left panel of the following figure. The true values of the coefficients are $\alpha = \beta = 1$. The LS estimates of α and β are 1.041 and 0.962, while their MAD estimates are 1.186 and 0.986, respectively.

The right panel shows the effects of two outliers (the last two observations are much smaller than those on the left panel). The LS estimates of α and β are 1.406 and 0.737, while their MAD estimates are 1.192 and 0.980, respectively. As expected, the two outliers have a substantial effect on the LS estimate of the slope, while they have almost no effects on its MAD estimate.



(b) M Estimators

Objective functions of the least squares estimator (LSE) and the MAD estimator can be considered as the weighted sum of the absolute values of the errors $|Y_i - \alpha - \beta X_i|$, where the weights are equal to the absolute error itself for the LSE and equal to 1 for the MAD estimator. Huber (1981) proposed alternative weight

functions to minimize the effects of the outliers. This class of estimators is called *M* estimators and minimizes

$$\min_{\alpha, \beta} \sum_{i=1}^n \psi(|Y_i - \alpha - \beta X_i|) |Y_i - \alpha - \beta X_i|$$

where $\psi(z)$ is a weight function of the absolute values of the errors. A weight function popularized by Huber is a linear function near zero and a constant function beyond a certain point k :

$$\psi(z) = \begin{cases} z & \text{if } z \leq k \\ k & \text{if } z > k \end{cases}$$

where $z = |Y_i - \alpha - \beta X_i|$. Note that the LSE is a special case of the *M* estimator with $k = \infty$, and the MAD estimator is a special case of a constant weight $\psi(z) = 1$.

Statistical Appendix

Multivariate Normal Random Vector

Definition: An n -dimensional random vector y is distributed as a multivariate normal, $y \sim N(\mu, \Sigma)$, with a mean vector μ and a nonsingular (positive definite) variance-covariance matrix Σ , if the joint density function is given by

$$f(y) = (2\pi)^{-n/2} |\Sigma|^{-1/2} \exp\left\{-\frac{1}{2}(y-\mu)' \Sigma^{-1}(y-\mu)\right\}.$$

Conditional Distribution

Theorem 1: Let y be distributed as $N(\mu, \Sigma)$. Partition y , μ , and Σ into

$$y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \quad \mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} \quad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

- (a) y_1 and y_2 are statistically independent if and only if $\Sigma_{12} = 0$.
- (b) The conditional distribution of y_1 given y_2 is normal:

$$y_1|y_2 \sim N\left\{\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(y_2 - \mu_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\right\}$$

- (c) y_2 and $(y_1 - \Sigma_{12}\Sigma_{22}^{-1}y_2)$ are statistically independent.

Distribution of a Linear Function

Theorem 2: Let y be an n -dimensional normal variate: $y \sim N(\mu, \Sigma)$. Then,

- (a) $(Ay - b) \sim N(A\mu - b, A\Sigma A')$, where A is an $m \times n$ matrix of constants with $\text{rank}(A) = m \leq n$, and b is an m -dimensional vector of constants.
- (b) Let $P'P = \Sigma^{-1}$. There exists such a nonsingular matrix P for a positive definite matrix Σ . Then, $P(y - \mu)$ is distributed as a standard multivariate normal: $N(0, I)$

Distribution of a Quadratic Function

Theorem 3: Let y be an n -dimensional normal variate: $y \sim N(\mu, \Sigma)$. Then,

- (a) $(y - \mu)' \Sigma^{-1}(y - \mu)$ is distributed as $\chi^2(n)$, a central chi-square random variable with n degrees of freedom.

Note: From the Theorem 2 (ii), we have $z = P(y - \mu) \sim N(0, I)$, that is, z_i are i.i.d. standard normal random variables. The quadratic function can be written as

$$(y - \mu)' \Sigma^{-1}(y - \mu) = (y - \mu)' P' P (y - \mu) = z' z = \sum_{i=1}^n z_i^2$$

which is the sum of squared independent standard normal random variables. Thus, we can define a χ^2 random variable with n degrees of freedom as the sum of n squared independent standard normal random variables

(b) $y' \Sigma^{-1} y$ is distributed as $\chi^2(n; \delta)$, a noncentral chi-square distribution with n degrees of freedom and the noncentrality parameter δ .

Note: Scheffe writes the noncentrality parameter as $\delta = (\mu' \Sigma^{-1} \mu)^{1/2}$. Some other authors use the notation $\lambda = \mu' \Sigma^{-1} \mu = \delta^2$ (see Rao), or $\lambda = \mu' \Sigma^{-1} \mu / 2 = \delta^2 / 2$ (see Graybill) for the noncentrality parameter. GAUSS follows Scheffe's notation in its noncentral cdf's (CDFTNC, CDFTNC, CDFENC).

(c) $(y - b)' \Sigma^{-1} (y - b)$ is distributed as $\chi^2(n; \delta)$, where $\delta^2 = (\mu - b)' \Sigma^{-1} (\mu - b)$ for GAUSS.

(d) If $\Sigma = \sigma^2 I$, then $y' A y / \sigma^2$ is distributed as $\chi^2(m, \delta)$ if and only if A is an *idempotent matrix* of rank m , where the noncentrality parameter is $\delta^2 = \mu' A \mu / \sigma^2$.

(e) Let y be an n -dimensional (not necessarily normal) random vector with $E(y) = \mu$ and $cov(y) = \Sigma$. Then, $E(y' A y) = tr(A \Sigma) + \mu' A \mu$. In particular, if $\Sigma = \sigma^2 I$, $\mu = 0$, and A is idempotent of rank m , then $E(y' A y) = \sigma^2 m$.

Independence of a Linear Function (Normal) and a Quadratic Function (Chi-Square)

Theorem 4. Let $y \sim N(\mu, \Sigma)$. Then,

(a) A linear function By and a quadratic function $y' A y$ are independent if $B \Sigma A = 0$, where A is a symmetric matrix.

(b) Two quadratic functions $y' A y$ and $y' B y$ are independent if $A \Sigma B = 0$.

F distribution and Student's t Distribution

Theorem 5. Let $Y \sim N(\mu, 1)$, $Z \sim \chi^2(n, \delta)$ and $W \sim \chi^2(m)$ which are independent. Then,

(a) $\frac{Y}{\sqrt{W/m}} \sim t(m; \mu)$ noncentral t distribution

(b) $\frac{Z/n}{W/m} \sim F(n, m; \delta)$, noncentral F distribution

Reproductive property of the independent noncentral χ^2

Theorem 6. Let X_i , $i=1, 2, \dots, n$ be mutually independent chi-square variates with v_i degrees of freedom and noncentrality parameter δ_i . Then,

$$\sum_{i=1}^n X_i \sim \chi^2(v, \delta), \quad v = \sum_{i=1}^n v_i, \quad \delta = \sum_{i=1}^n \delta_i$$

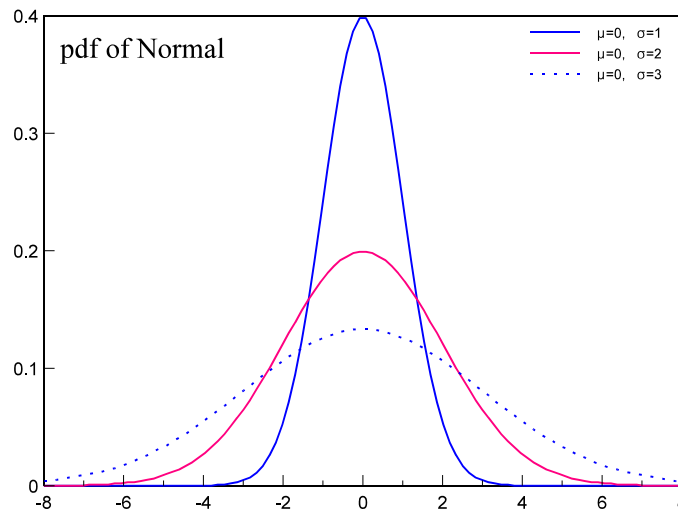
Independence of measurable functions of random variables

Theorem 7. If X and Y are stochastically independent, then any measurable functions $Z = u(X)$ and $W = v(Y)$ are also independent.

Special Continuous Distribution Functions

Normal distribution $X \sim N(\mu, \sigma^2)$. This distribution with two parameters, μ and σ , and its pdf is

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}, \quad -\infty < x < +\infty, \quad -\infty < \mu < +\infty, \quad \sigma > 0$$



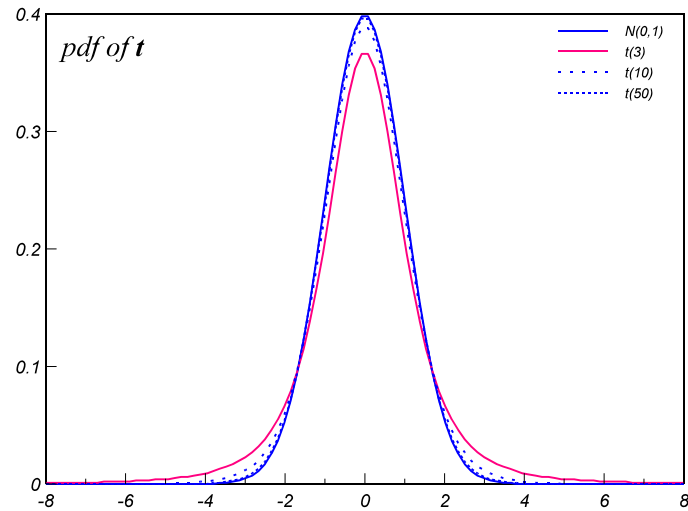
Central Student's t distribution. The central t distribution arises as the distribution of a ratio of a standard normal random variable $N(0,1)$ and a central chi-square random variable with m degrees of freedom. The parameter $m > 0$ is also called the degrees of freedom of the t distribution. The pdf is

$$f(x; m) = \frac{\Gamma((m+1)/2)}{\Gamma(m/2)\sqrt{\pi m}} \frac{1}{(1 + x^2/m)^{(m+1)/2}} \quad -\infty < x < \infty$$

which has $E(X)=0$ for $m > 1$ and $\text{var}(X)=m/(m-2)$ for $m > 2$. This density is symmetric around 0 and looks very much like a normal pdf except that it has thicker tails than a normal pdf. Let $Z \sim N(0,1)$ and $Y \sim \chi^2(m)$. If they are independent,

$$X = \frac{Z}{\sqrt{Y/m}}$$

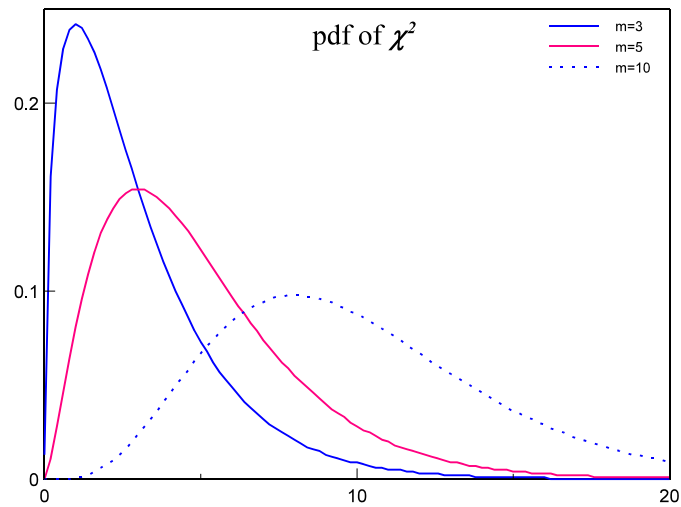
has t -distribution with m degrees of freedom.



Central chi-square distribution. This is denoted by $\chi^2(m)$ or χ_m^2 . The parameter m , which is generally a positive integer, is called *the degrees of freedom*.

$$f(x;m) = \frac{1}{\Gamma(m/2)2^{m/2}} x^{(m-2)/2} e^{-x/2}, \quad x \geq 0$$

$$E(X) = m, \quad \text{var}(X) = 2m$$



Central F-distribution. This is denoted by $F(m,n)$, where m and n are positive integers and they are called the degrees of freedom. This distribution is derived from a ratio of independent chi-square distributions $Z \sim \chi^2(m)$, $Y \sim \chi^2(n)$, and Z and Y be independent

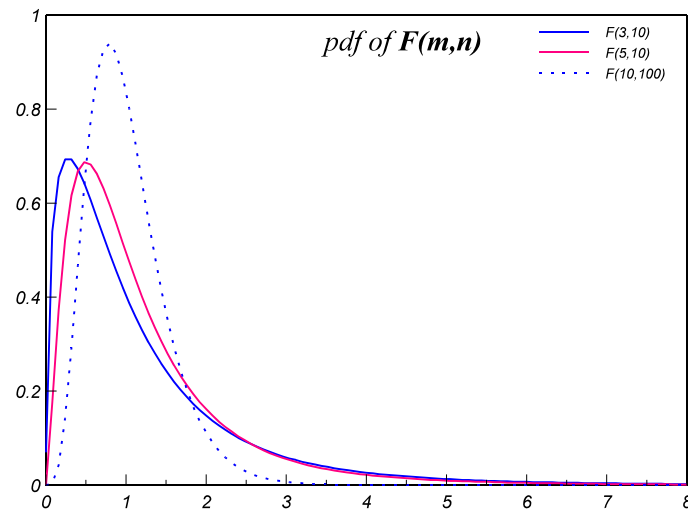
$$X = \frac{Z/m}{Y/n} \sim F(m, n)$$

The density function is

$$f(x) = \frac{c^{m/2} x^{(m-2)/2}}{\Gamma(m/2)\Gamma(n/2) (1+cx)^{(m+n)/2}} \Gamma((m+n)/2), \quad c = \frac{m}{n}$$

$$E(X) = \frac{n}{n-2} \quad \text{for } n > 2$$

$$\text{var}(X) = \frac{2n^2(m+n-2)}{m(n-2)^2(n-4)} \quad \text{for } n > 4$$

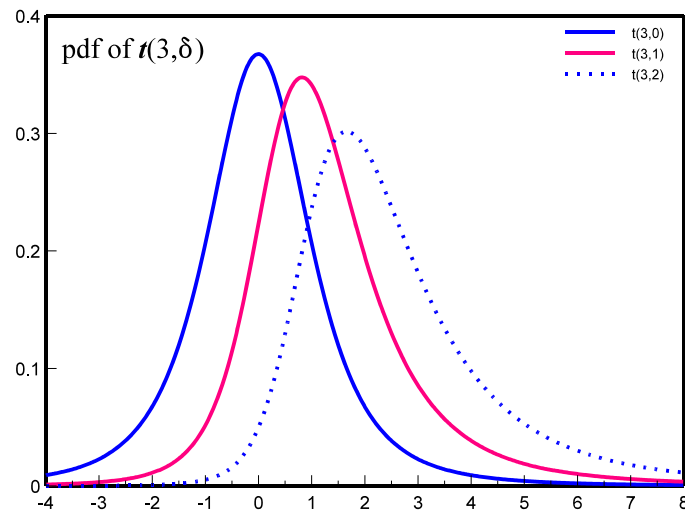


Noncentral t -distribution:

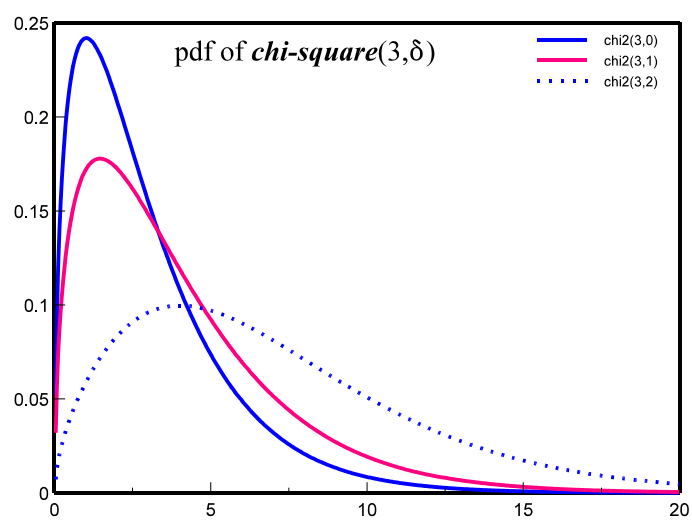
Let $Z \sim N(\mu, 1)$ and $W \sim \chi^2(n)$, and Z and W are independent. Then, $\frac{Z}{\sqrt{W/n}} \sim t(n, \mu)$ is distributed as a

noncentral t distribution with degrees of freedom n and the noncentrality parameter μ which can be positive or negative. Its density function is

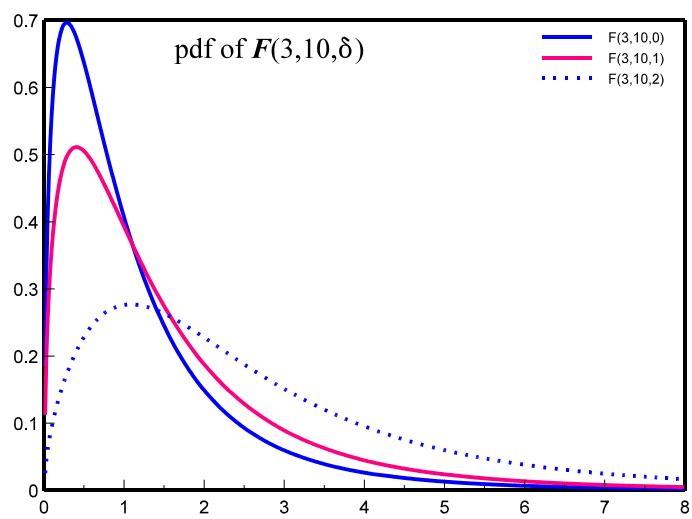
$$\begin{aligned} f(x; n, \mu) &= \frac{e^{-\mu^2/2}}{\sqrt{\pi n} \Gamma(n/2)} \left(\frac{n}{n+x^2} \right)^{(n+1)/2} \sum_{j=0}^{\infty} \frac{1}{j!} \Gamma\left(\frac{n+1+j}{2}\right) \left(\frac{x\mu\sqrt{2}}{\sqrt{n+x^2}} \right)^j \\ &= \frac{e^{-\mu^2/2}}{\sqrt{\pi} \Gamma(n/2)} \sum_{j=0}^{\infty} \frac{\Gamma((n+1+j)/2)}{j! v^{(j+1)/2}} (\mu \sqrt{2} x)^j \left(1 + \frac{x^2}{v} \right)^{-(v+1+j)/2} \end{aligned}$$



Noncentral χ^2 distribution



Noncentral F-distribution



Matrix Algebra

Theorem. Partitioned Matrix. Let A be an n-dimensional square matrix and partition A into

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$$

Define

$$B = A_{11} - A_{12}A_{22}^{-1}A_{21} \quad \text{if } |A_{22}| \neq 0$$

$$C = A_{22} - A_{21}A_{11}^{-1}A_{12} \quad \text{if } |A_{11}| \neq 0$$

Then,

$$|A| = \begin{cases} |B| |A_{22}| & \text{if } |A_{22}| \neq 0 \\ |C| |A_{11}| & \text{if } |A_{11}| \neq 0 \end{cases}$$

If A is nonsingular and inverses of the matrices defined below exist, then we can write the inverse of a partitioned matrix as

$$A^{-1} = \begin{pmatrix} B^{-1} & -B^{-1}A_{12}A_{22}^{-1} \\ -A_{22}^{-1}A_{21}B^{-1} & A_{22}^{-1} + A_{22}^{-1}A_{21}B^{-1}A_{12}A_{22}^{-1} \end{pmatrix} = \begin{pmatrix} A_{11}^{-1} + A_{11}^{-1}A_{12}C^{-1}A_{21}A_{11}^{-1} & -A_{11}^{-1}A_{12}C^{-1} \\ -C^{-1}A_{21}A_{11}^{-1} & C^{-1} \end{pmatrix}$$

Matrix Differential Calculus

Definition: Matrix Differential - Let X be an m×n matrix. The matrix differential dX is an m×n matrix defined by dX=(dx_{ij}).

Definition: Scalar Function - Let f(X) be a scalar function of an m×n matrix X. Then, df/dX is an m×n matrix defined by the element wise derivative df(X)/dX=(df/dx_{ij}).

Definition: Vector Function - Let y=f(x) be an n×1 column vector function of an m×1 column vector x. Then, dy/dx=(dy_j/dx_i) is an m×n matrix, where dy_j/dx_i is the ijth element of dy/dx.

$$y = f(x), \quad y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix}, \quad \frac{dy}{dx} = \begin{pmatrix} \frac{dy_1}{dx_1} & \frac{dy_2}{dx_1} & \dots & \frac{dy_n}{dx_1} \\ \frac{dy_1}{dx_2} & \frac{dy_2}{dx_2} & \dots & \frac{dy_n}{dx_2} \\ \vdots & \vdots & \dots & \vdots \\ \frac{dy_1}{dx_m} & \frac{dy_2}{dx_m} & \dots & \frac{dy_n}{dx_m} \end{pmatrix}$$

In particular, if y is a scalar function of vector x (i.e., n=1), then dy/dx is an m×1 column vector of element

wise derivative $dy/dx=(dy/dx_i)$.

Theorem: Linear Function - Let $f(X)$ be a *linear* function of X . Then, $df(X)=f(dX)$.

Example. $d\text{tr}(X)=\text{tr}(dX)$. $d(\text{vec}X)=\text{vec}(dX)$. $d(Ax)=Adx$.

Theorem: Let $y=f(x)$ be a scalar-valued or a column vector function of a column vector x . Then, $dy=Adx$ if and only if $dy/dx=A'$.

Example. Let $y=Ax$. Then, $dy=Adx$, and hence, $dy/dx=A'$.

Theorem: Let X and Y be matrices conformable for multiplication. Then, $d(XY)=(dX)Y+X(dY)$, and $d(XAY)=(dX)AY+XA(dY)$ for a constant matrix A .

Example. Let $y=x'Ax$ be a scalar quadratic function of a vector x . Then,

$$dy = (dx')Ax + x'A(dx) = x'A'(dx) + x'A(dx) = x'(A'+A)(dx)$$

where we used the fact that $(dx')Ax$ is a scalar and hence it is equal to its transpose $x'A'(dx)$. Therefore, $dy/dx = (A'+A)x$. In particular, $d(x'x)/dx=2x$, and $d(x'Ax)/dx=2Ax$ if A is symmetric.

Example. Let $z=(y-X\beta)'(y-X\beta)$, where y and X are, respectively, $n \times 1$ vector and $n \times K$ matrix, and β is a $K \times 1$ vector. We wish to find $\partial z/\partial \beta$. Expand the multiplication into

$$z = (y-X\beta)'(y-X\beta) = y'y - y'X\beta - \beta'X'y + \beta'X'X\beta$$

Treating y and X as constant matrices, we have

$$\begin{aligned} dz &= -y'Xd\beta - (d\beta')X'y + (d\beta')X'X\beta + \beta'X'Xd\beta \\ &= -y'Xd\beta - y'Xd\beta + \beta'X'Xd\beta + \beta'X'Xd\beta = (-2y'X + 2\beta'X'X)d\beta \end{aligned}$$

which gives

$$\partial z/\partial \beta = -2X'y + 2X'X\beta.$$

The second order derivative is easily seen to be $\partial^2 z/\partial \beta \partial \beta' = 2X'X$.